
Commutative Algebra

— *EYNTKA* Series —

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Contents

Contents	3
I Commutative Algebra	5
1 More on Ring Theory, Ring Extensions, and Category Theory	9
2 Categorical (co)Limits, Natural Transformations, and Adjoint	11
3 Noetherian Rings, Hilbert's Basis Theorem, and Localization	23
3.1 Hilbert Basis Theorem	25
3.2 Gröbner Basis	29
3.3 Localization	29
3.3.1 Localization of Rings	29
3.3.2 Localization of Modules	36
3.3.3 Local Properties	38
3.3.4 Local Rings	41
3.3.5 Nakayama's Lemma	43
4 Completion	51
4.1 Constructing Completions	52
4.2 Filtrations	59
4.3 More on Graded Algebras	60
4.4 Artin-Rees Lemma	62

4.5	Associated Graded Rings: Noetherian Closed under Completion	66
5	Ring Extension and Integral Extension	71
5.1	Extension and Contraction of Ideals	76
5.2	Integral Extensions and Prime Ideals	79
6	Valuation Ring and Local Rings	81
6.0.1	Discrete Valuation Ring	84
7	Artinian Rings	87

Part I

Commutative Algebra

In this part of the book, we are no longer studying a new object as the central focus. Rather, it is the basic language that deals with *algebraic number theory* and *algebraic geometry*; this is the language of commutative algebra. Commutative algebra is an extension of ring theory where it is always assumed that the ring R is commutative. As shall be shown soon, this theory will prove to be very useful in developing two major studies of mathematics:

1. *Algebraic Geometry*: At the heart of Algebraic Geometry is the realization that all commutative rings have a corresponding *geometric object* (known as an *affine scheme*). We have already encountered the simplest of schemes in section ?? with the classification of hyper-planes of a finitely dimensional vector-space. Recall that each hyper-plane is associated the solution (or kernel) of linear equations. Schemes generalize this notion a great deal by working with polynomials over general commutative rings instead of restricting to linear equations (which are polynomials of degree 1), and by allowing the resulting geometric object to be “glued together” from kernels of polynomials instead of being the solution to polynomials. In chapter ??, we shall explore how to associate a scheme to every commutative ring and build-up the interesting theory behind this correspondence.
2. *Algebraic Number Theory*: Studying fields allowed us to study the solutions to polynomials over fields. However, fields are very restrictive (dually very versatile) algebraic structures that allowed for some very powerful results. When trying to replicate the same strategies for solving polynomial equations over more general rings, many of these strategies fall apart. To properly generalize this, we shall need to invent a new theory, namely *algebraic number theory* is exactly this theory. In chapter ??, we shall explore some results such as quadratic reciprocity and classifying all Abelian extensions (i.e. Kronecker-Weber’s Theorem). Most of the time, we focus on solutions to equation over $\mathbb{Z}$, as $\mathbb{Z}$ is the initial object of ring, or more concretely $\mathbb{Z}$ is an object that appears very naturally in the world.

In this chapter, some basic topological notions will be used. The geometric intuition presented in this chapter will be lost on those who have not taken or learned some basic topology (up to the separation axioms, in particular T_1 and T_2 , and compactness/connectedness). Appendix ?? serves as a review of what topological concepts will be used in this chapter.

Chapter ?? and ?? can be thought of as covering the more “classical” versions of their respective fields. For example, modern algebraic geometry shall start with schemes, while algebraic number theory is developed over local fields and uses techniques from complex analysis. Most of the motivation for these developments come from more concrete questions that naturally arose when studying polynomials and their solutions. The reader shall certainly gain a lot by understanding the core ideas behind these two chapters before tackling their more abstract modernizations. The reader wishing to follow up these chapters with a more in-depth covering can checkout [NateAlgGeo] for the continuation of this book towards Algebraic Geometry and [NateCFT] for a continuation of this book towards Class Field Theory.

Note that algebraic geometry is *not* algebraic topology. In algebraic topology, we usually start with a topological space (or manifold) and associate to it an algebraic object to find properties of the topological/geometrical object. In Algebraic Geometry, we start with a commutative ring (or a collection of commutative rings), and create a geometric object that can be used to study the ring.

1

***More on Ring Theory, Ring
Extensions, and Category Theory***

(this chapter must be deleted once the label is safely replaced)

2

Categorical (co)Limits, Natural Transformations, and Adjoints

In this chapter, it will be taken that chapter ?? has been read, the rigorous definition of category, functor, and commutative diagram are all well understood, and simple examples of all of these concepts come to mind. Recall that a functor could be covariant or contravariant. To demonstrate, if we have a functor $F : C \rightarrow D$ between two categories and a commutative diagram in the category C as follows:

$$\begin{array}{ccccc} b & \longrightarrow & c & \longrightarrow & f \\ \uparrow & \searrow & \uparrow & \swarrow & \uparrow \\ a & \longrightarrow & d & \longrightarrow & e \end{array}$$

Then if F is covariant, the above commutative diagram gets sent to:

$$\begin{array}{ccccc} F(b) & \longrightarrow & F(c) & \longrightarrow & F(f) \\ \uparrow & \searrow & \uparrow & \swarrow & \uparrow \\ F(a) & \longrightarrow & F(d) & \longrightarrow & F(e) \end{array}$$

and if F is contravariant, the commutative diagram gets sent to:

$$\begin{array}{ccccc} F(b) & \longleftarrow & F(c) & \longleftarrow & F(f) \\ \downarrow & \swarrow & \downarrow & \searrow & \downarrow \\ F(a) & \longleftarrow & F(d) & \longleftarrow & F(e) \end{array}$$

With the further material covered, a couple more interesting functors may be introduced which may enlighten the reader with some further intuitions on functors:

Example 2.0.1: Further Examples Of Functors

1. Ring and Modules both have natural forgetful functors to either **Set** or **Ab**. The adjoint of the forgetful functor $\mathbf{Ring} \rightarrow \mathbf{Ab}$ maps abelian groups to group-rings.
2. Let R be a ring and R^* be its group of units. Then any ring homomorphism $R \rightarrow S$ can be induces a group homomorphism $R^* \rightarrow S^*$ (namely since ring homomorphisms sends units to units).
3. Let S be any set. Take $\hat{S}$ to be the category whose objects are subsets of S and morphisms are inclusions $\subseteq$. Then a contravariant functor $\hat{S} \rightarrow \mathbf{Set}$ is called a *presheaf*. A typical example would associate to each $U \subseteq S$ the set of all function $\text{Hom}_C(S, X)$ for some X (popular examples are $\mathbb{R}, \mathbb{C}, \mathbb{Z}, S^1$, though there are many others) and locally small category C^a , and each $U \subseteq V$ is associated to a restriction map $\text{Hom}_C(V, X) \rightarrow \text{Hom}_C(U, X)$, $\varphi \mapsto \varphi|_U$. These types of functors are very important and shall be further explore in algebraic geometry.
4. As a particular example of the above, let C be any locally small category and $X \in C$. Then:

$$A \mapsto \text{Hom}_C(X, A) \quad A \mapsto \text{Hom}_C(A, X)$$

define a covariant and contravariant functor $C \rightarrow \mathbf{Set}$ respectively; these functors are usually denoted $\text{Hom}_C(-, A)$ and $\text{Hom}_C(A, -)$. To see where the morphisms of C get mapped to, it's best to see the following diagrams:

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & C \\ & \searrow f_a & \downarrow f_b & \swarrow f_c & \\ & & X & & \end{array} \quad \begin{array}{ccc} & & X & & \\ f_a \swarrow & & \downarrow f_b & & \searrow f_c \\ A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & C \end{array}$$

Taking $\text{Hom}_C(-, X)$, any morphism $\alpha : A \rightarrow B$ gets mapped to $\text{Hom}_C(B, X) \rightarrow \text{Hom}_C(A, X)$ where $f_b \mapsto f_a = (f_b \circ \alpha)$. The functors $\text{Hom}_C(-, X)$ and $\text{Hom}_C(X, -)$ are often abbreviated to h_X and h^X respectively.

5. Recall that if $C = R\text{-Mod}$, then $\text{Hom}_R(A, B)$ is an abelian group. If a functor F preserves this structure, then it is called *additive*. In this book, F would usually be a functor between categories of modules. For example, $F : R\text{-Mod} \rightarrow S\text{-Mod}$ is an additive functor then:

$$\text{Hom}_R(A, B) \rightarrow \text{Hom}_S(F(A), F(B))$$

is a group homomorphism. for all $A, B \in \text{Obj}(R\text{-Mod})$. As an example: $S \otimes_R (-) : R\text{-Mod} \rightarrow S\text{-Mod}$ is an additive functor mapping $M \mapsto S \otimes_R M$ and $f : M \rightarrow N$ to $\iota \otimes f : S \otimes_R M \rightarrow S \otimes_R N$.

^arecall that a locally small category is one where $\text{Hom}_C(A, B)$ has at most a set amount of elements

Natural Transformations and Equivalence of Categories

Continuinig from chapter ??, it is important to establish when two categories are equivalent, that is a notion of “isomorphism of categories”. If $F : C \rightarrow D$ were an isomorphism, Then there would exists

a G such that $F \circ G = G \circ F = \text{id}$. However, in general this definition is too restrictive to be of much use. This comes down to the fact that most objects in category theory are defined *up to isomorphism*¹. In most cases, what we really need is that $F \circ G$ and $G \circ F$ are both “isomorphic” to id . In order to talk about functors being isomorphic, we must talk about morphisms between functors. This might seem like rather abstract concept. However, they have a very practical purpose: they represent when two “interpretations” provided by a functor are comparable *without* any supplementary information. This was briefly commented upon this notion in section ?? when discussing how when V is finite dimensional, $V \cong V^*$ and $V \cong V^{**}$. The isomorphism $V \cong V^*$ required a choice of basis, while $V \cong V^{**}$ can be done without the need of any such choice, hence making the later more *natural*. What this translates to is the function $(-)^* : \mathbf{Vect}_{\text{fin}} \rightarrow \mathbf{Vect}_{\text{fin}}$ is *not* naturally isomorphic to $\text{id} : \mathbf{Vect}_{\text{fin}} \rightarrow \mathbf{Vect}_{\text{fin}}$, while $(-)^{**} : \mathbf{Vect}_{\text{fin}} \rightarrow \mathbf{Vect}_{\text{fin}}$ is naturally isomorphic to id . The following definition captures this idea:

Definition 2.0.2: Natural Transformation

Let $F : C \rightarrow D$ and $G : C \rightarrow D$ be two functors between the same categories. Then a *natural transformation* from F to G , denoted $\alpha : F \Rightarrow G$ ^a consists of a morphism $\alpha_c \in \text{Mor} D$ for each $c \in \text{Obj}(C)$, where $\alpha_c : F(c) \rightarrow G(c)$ and that for any morphism $f : c \rightarrow c' \in \text{Mor} C$, the following diagram commutes.

$$\begin{array}{ccc} F(c) & \xrightarrow{\alpha_c} & G(c) \\ F(f) \downarrow & & \downarrow G(f) \\ F(c') & \xrightarrow{\alpha_{c'}} & G(c') \end{array}$$

A *natural isomorphism* is a natural transformation α such that each α_c is an isomorphism, in which case we often write $\alpha : F \cong G$.

^athe notation $\Rightarrow$ instead of $\rightarrow$ is due to historical precedence; it has no relation to the logical notion of implication

(couple of interesting examples that are currently in the category chapter of the book.)

Natural transformation shall be covered in more depth in part ??. At the moment, they are useful to define the following weaker definition of equivalence of categories

Definition 2.0.3: Equivalence Of Categories

Let C and D be two categories. Then C and D are said to be *equivalent* if there exists functors $F : C \rightarrow D$ and $G : D \rightarrow C$ such that $F \circ G \cong 1_D$ and $G \circ F \cong 1_C$, that is $F \circ G$ and $G \circ F$ are both naturally isomorphism to the identity functor. If C and D are naturally equivalent, then we shall write $C \cong D$

It should be verified that equivalence of categories defines an equivalence relation. This definition is nice theoretically, but for practical purposes, the following equivalent definition is what will be used instead

¹There will be examples of isomorphism of categories, famously the Galois Correspondence can be made into a category isomorphism between the right categories

Proposition 2.0.4: Equivalence Of Categories

Let C and D be categories. Then $C \cong D$ if and only if there exists a functor F such that

1. F is a *full* functors, that is for each $x, y \in C$, $\text{Hom}_C(x, y) \rightarrow \text{Hom}_D(F(x), F(y))$ is *surjective*
2. F is a *faithful* functors, that is for each $x, y \in C$, $\text{Hom}_C(x, y) \rightarrow \text{Hom}_D(F(x), F(y))$ is *injective*
3. F is an *essentially surjective* functors, that is for each $y \in D$, there exists a $c \in C$ such that $F(c) \cong d$

Proof:

see ref:HERE in part ??.

Example 2.0.5: Equivalence Of Categories

Let $\mathbf{Mat}_k$ be the categories whose objects are the natural numbers $\mathbb{N}$ and whose morphisms between any two numbers, $\text{Hom}(m, n)$, is the set of $n \times m$ matrices with entries in the field k , where composition is defined as the product of matrices. Then $\mathbf{Mat}_k \cong \mathbf{Vect}_k^{\text{fin}}$, namely each n gets mapped to k^n with the standard basis for k^n , and each matrix gets mapped to the linear transformation corresponding to the matrix given the standard basis. It is easy to see that this functor is fully faithful and essentially surjective, making these categories equivalent.

Limits and Colimits

The next important concepts is that of *limits and colimits*. If the reader knows some analysis, they might suspect there to be some similarities between these concepts, and abstractly there are, as shall be demonstrated. On a high-level, a limit and colimit is always taken given a commutative diagram, say for simplicity we have the diagram:

$$a \rightarrow b \rightarrow c$$

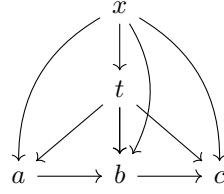
Focusing for a moment on limits, we may imagine that there are multiple possible morphism from an object, say x , to objects a, b, c such that the following diagram commutes:

$$\begin{array}{ccccc} & & x & & \\ & \swarrow & \downarrow & \searrow & \\ a & \longrightarrow & b & \longrightarrow & c \end{array}$$

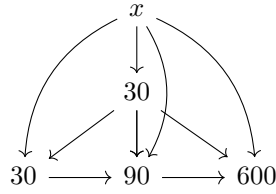
For example, if a, b, c where natural numbers and the morphism $\alpha \rightarrow \beta$ exists if and only if $\alpha | \beta$, then x would have to divide a, b, c , then we may have the following commutative diagrams:

$$\begin{array}{ccccc} & & 10 & & \\ & \swarrow & \downarrow & \searrow & \\ 30 & \longrightarrow & 90 & \longrightarrow & 600 \end{array} \quad \begin{array}{ccccc} & & 2 & & \\ & \swarrow & \downarrow & \searrow & \\ 30 & \longrightarrow & 90 & \longrightarrow & 600 \end{array} \quad \begin{array}{ccccc} & & 5 & & \\ & \swarrow & \downarrow & \searrow & \\ 30 & \longrightarrow & 90 & \longrightarrow & 600 \end{array}$$

Then the limit of this diagram, say t is the “terminal” object of the above, that is for any x such that the diagram commutes:



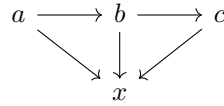
For example, in the above diagram, 30 is the terminal object, since: for any other number x such that $x \rightarrow 30$, $x \rightarrow 90$, and $x \rightarrow 600$ (that is $x|30$, $x|90$, and $x|600$), then necessarily $x \rightarrow 30$, that is $x|30$,



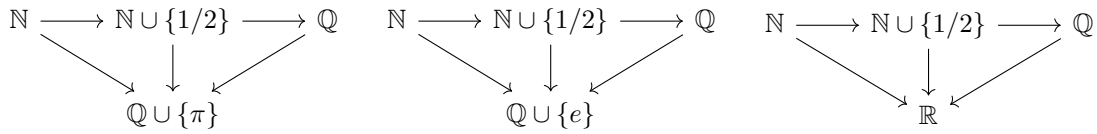
Similar to how a limit in calculus and analysis is in some way the “end” of a sequence, the limit of a diagram is the “end” of a diagram. Dually, a *colimit* reverses the process. Just like limits, any colimit start with a given commutative diagram, say the same commutative diagram that was used above:

$$a \rightarrow b \rightarrow c$$

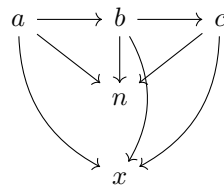
Then given the above commutative diagram, there may be many morphisms from a, b, c to an object, say x , the following diagram always commutes:



For example, if a, b, c where sets and a morphism $\alpha \rightarrow \beta$ exists if and only if $\alpha \subseteq \beta$, then we may have the following commutative diagrams:



then the colimit of this diagram, say n , is the “initial” object of the above, that is for any x , the following diagram always commutes:



For example, in the above diagram the colimit is $\mathbb{Q}$. To generalize this notion, the notion of commutative diagram must be abstracted:

Definition 2.0.6: Commutative Diagram

Let C be a category. Then a *diagram* in C is a function $F : I \rightarrow C$ whose domain is an *indexing category* which is a small category.

An indexing category is an arbitrary small category which consists of arbitrary objects and morphisms. The category is usually finite or countable and has very few morphisms. Omitting the identity morphisms, an indexing category may look like any of the following when all the objects and morphisms are drawn out as a directed graph:

1. $\bullet \longrightarrow \bullet \longrightarrow \bullet$
2. $\begin{array}{cc} \bullet & \bullet \end{array}$
3. $\begin{array}{ccc} \bullet & \longleftarrow & \bullet \\ \downarrow & \nearrow & \uparrow \\ \bullet & \longrightarrow & \bullet \end{array}$
4. $\bullet \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \dots$

Definition 2.0.7: [Categorical] Limits

Let $\mathbf{C}$ be a category, $\mathbf{I}$ be an indexing category, and $F : \mathbf{I} \rightarrow \mathbf{C}$ a covariant functor. Then the *limit* of F (if it exists) is an object $L \in \mathbf{C}$ endowed with morphisms $\varphi_I : L \rightarrow F(I)$ for all object $I \in \mathbf{I}$ such that

1. If $\alpha : I \rightarrow J$ is a morphism in $\mathbf{I}$, then $\varphi_J = F(\alpha) \circ \varphi_I$, that is:

$$\begin{array}{ccc} L & & \\ \downarrow \varphi_I & \searrow \varphi_J & \\ F(I) & \xrightarrow{F(\alpha)} & F(J) \end{array}$$

2. L is final with respect to this property that is, if M is another object endowed with morphism ψ_I that also satisfies the previous requirement, then there exists a unique morphism $M \rightarrow L$ making the relevant diagrams commute

the limit is usually denoted $\varprojlim F$. A limit is sometimes also called an *inverse limit* or *projective limit*.

dually:

Definition 2.0.8: [Categorical] Colimit

Let $\mathbf{C}$ be a category, $\mathbf{I}$ be an indexing category, and $F : \mathbf{I} \rightarrow \mathbf{C}$ a covariant functor. Then the *limit* of F (if it exists) is an object $L \in \mathbf{C}$ endowed with morphisms $\varphi_I : L \rightarrow F(I)$ for all object $I \in \mathbf{I}$ such that

1. If $\alpha : J \rightarrow I$ is a morphism in $\mathbf{I}$, then $\varphi_I = \varphi_J \circ F(\alpha)$, that is:

$$\begin{array}{ccc} L & & \\ \downarrow \varphi_I & \searrow \varphi_J & \\ F(I) & \xrightarrow{F(\alpha)} & F(J) \end{array}$$

2. L is initial with respect to this property that is, if M is another object endowed with morphism ψ_I that also satisfies the previous requirement, then there exists a unique morphism $L \rightarrow M$ making the relevant diagrams commute

the limit is usually denoted $\varinjlim F$. A colimit is sometimes also called a *direct limit* or an *injective limit*.

Example 2.0.9: Limits and Colimits

It should be stated immediately that all universal properties are examples of limits and colimits, and the reader should take these for their basic understanding of (co)limits. We shall take most of our examples in **CRing** and **R-Mod** for commutative ring R , as it is the category most relevant to the following chapters and in algebraic geometry.

1. The simplest diagram to take the limit and colimit of is the diagram with a single dot:

•

In **CRing**, the Limit and Colimit will just return the ring itself

• •

The limit of this diagram is called the *product*, and the colimit the *coproduct*. In **CRing**, the product and coproduct match, that is:

$$\begin{array}{ccc} & R \times S & \\ \swarrow \pi_R & & \searrow \pi_S \\ R & & S \end{array} \quad \text{and} \quad \begin{array}{ccc} R & & S \\ \searrow \iota_R & & \swarrow \iota_S \\ & R \times S & \end{array}$$

Are the limit and colimits in **CRing**. In **Ring**, the colimit of a diagram would be the free product.

2. Let us now take a diagram with two arrows:

$$\begin{array}{ccc} \bullet & \xrightarrow{\quad} & \bullet \\ & \xrightarrow{\quad} & \end{array}$$

The two arrows usually represent that we want certain values from the domain of the arrow to be equal in the codomain. This diagram is called the *equalizer*. We would be a bit more

careful with which category to take the limit, as we usually want the limit to be a single arrow $K \rightarrow X \rightrightarrows Y$ to be the universal object. The quintessential example of a limit of such a diagram is with one of the arrows being the 0 morphism (when it exists in the category):

$$R \begin{array}{c} \xrightarrow{0} \\ \xrightarrow{f} \end{array} S$$

In **CRing**, the object that would satisfy this arrow would have to commute in the diagram, namely:

$$f(R) = 0(R) = 0$$

Namely, the *kernel*, and $\ker(f) \xrightarrow{f} R \rightrightarrows S$.

3. We may wonder which limits may exist in a category. One famous result called the *Existence Theorem for Limits* is that all limits exist in a category (it is *complete*) iff all equalizers and (small) products exists (similarly for all colimits existing with equalizers and coproducts, this makes the category *cocomplete*). A small product means that it is closed under a “set-size” of indexing, something that is not an issue for our current investigation as we shall always have a set-size for our indexing, usually finite countable, or $|\mathbb{R}|$. The category **CRing** and **R-Mod** both have all [small] (co)products and (co)equalizers, and hence they are both (co)complete.

A category that is not complete but not cocomplete is the category of compact Hausdorff spaces (or by Gelfand Duality^a, unital commutative C*-algebras) which is closed under countable product (by Tychonoff’s theorem and some more results from functional analysis^b) and closed under equalizers, but is closed under coproducts as this can result in a discrete topology on $\mathbb{Z}$. The category of finite groups is neither complete nor cocomplete. f

4. Consider the indexing category I which may be represented as the following diagram:

$$\cdots \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$$

Then $F : \mathbf{I} \rightarrow \mathbf{C}$ represents a coiceof objects A_i an morphism $\varphi_{ji} : A_i \rightarrow A_j$ such that $\varphi_{ii} = 1_{A_i}$ and $\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$ for all $i \geq j \geq k$ giving us the diagram:

$$\cdots \xrightarrow{\varphi_{45}} A_4 \xrightarrow{\varphi_{34}} A_3 \xrightarrow{\varphi_{23}} A_2 \xrightarrow{\varphi_{12}} A_1 \quad (2.1)$$

Then $\varprojlim F$ is an object A endowed with morphisms $\varphi_i : A \rightarrow A_i$ such that the following diagram commutes:

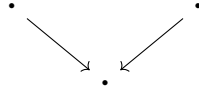
$$\begin{array}{ccccccc} & & & & A & & \\ & & \cdots & & \downarrow \varphi_3 & \searrow \varphi_2 & \searrow \varphi_1 \\ \cdots & \xleftarrow{\varphi_{45}} & A_4 & \xrightarrow{\varphi_{34}} & A_3 & \xrightarrow{\varphi_{23}} & A_2 \xrightarrow{\varphi_{12}} A_1 \end{array}$$

and A is terminal, that is any object object that satisfies the above diagram factors through A . Often, such a limit is denoted $\varprojlim_i A_i$. Such a limit need not always exist, however in **RMod**, we have that they always exist. Indeed, say we are given a diagram as in equation (2.1). Take $\prod_i A_i$, which we may interpret as a set of sequences. Take $A \subseteq \prod_i A_i$ to be all sequence $(a_i)_{i \geq 0}$ which are *coherent*, that is for all $i \geq 0$, take $a_i = \varphi_{i,i+1}(a_{i+1})$. The reader may verify that A is indeed a R -submodule of $\prod_i A_i$, and that the canonical projection $\pi_i : A \rightarrow A_i$

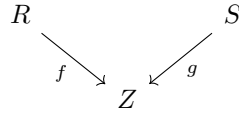
will make the necessary diagram commute. Then it is a relatively simple check to see that $A = \lim_i A_i$. a very similar construction occurs for colimits, in particular with the indexing category:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow \dots$$

5. An important generalization of the product and coproduct is the *fibred product* (or pullback) and *fibred coproduct* (or pushout). The fibred product is given by the diagram:



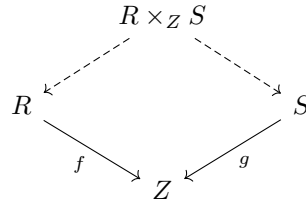
The intuition is that the left and right dot are somehow glued at the same place for the bottom product. In **CRing** and **R-Mod**, the fibred product of:



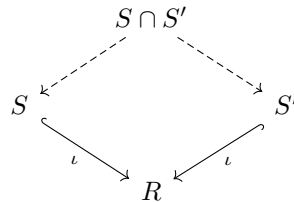
is:

$$R \times_Z S = \{(r, s) \in R \times S \mid f(r) = g(s)\}$$

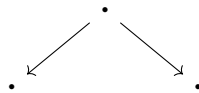
This is also called the pullback as it fits nicely into the diagram:



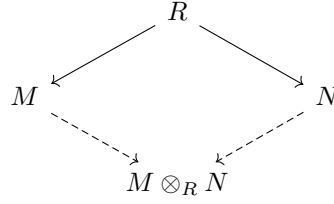
Notice that the intersection can be written as a fibred product, namely if $S, S' \subseteq R$, then:



The fibred coproduct is the colimit of the codiagram:



Examples of coproducts come from tensor products over a [commutative] ring R :



In **CRing**, we can consider R as $\mathbb{Z}$ -modules.

^asee eyntka cite:funcAnalysis

^bsee cite:HERE

Adjoint Functors

The final categorical that will be introduced is a special type of equivalence of functors that is a little moer general than natural transformations:

Definition 2.0.10: Adjoint Functors

Let $F : \mathbf{C} \rightarrow \mathbf{D}$ and $G : \mathbf{D} \rightarrow \mathbf{C}$ be two functors. Then these functors are *adjoint*, in particular F is *left-adjoint* to G and G is *right-adjoint* to F) if there is a natural isomorphism:

$$\mathrm{Hom}_C(X, G(Y)) \xrightarrow{\sim} \mathrm{Hom}_D(F(X), Y)$$

for all objects $X \in C$ and $Y \in D$.

If you are more proficient with category theory, we would say there is a natural isomorphism of *bifunctors* $C \times D \rightarrow \mathbf{Set} : \mathrm{Hom}_C(-, G(-)) \xrightarrow{\sim} \mathrm{Hom}_D(F(-), -)$. A functor F is called a *left-adjoint* functor (resp. *right-adjoint* functor) if there exists a corresponding right-adjoint functor (resp. left-adjoint functor).

Example 2.0.11: Adjoints

1. (The free-forgetful adjoint)
2. Any equivalence of categories naturally comes with a functor with an adjoint pair.
3. (adjoining an identity to a rng. Similarly wth adjoining identity to semi-groups, or building up from monoids to groups to rings)
4. (scalar extension is adjoint with respect to $\varphi : R \rightarrow S$ is left adjoint to the forgetful functor $G : S\text{-}\mathbf{Mod} \rightarrow R\text{-}\mathbf{Mod}$)
5. (Field of fractions has an adjoint. In particular the category $\mathbf{Dom}_m$ whose objects are integral domains and morphism are injective ring homomorphisms. Then the forgetful functor $\mathbf{Fld} \rightarrow \mathbf{Dom}_m$ is right-adjoint to the functor $\mathbf{Dom}_m \rightarrow \mathbf{Fld}$ that assigns each element of the domain its field of fraction)

6. (The inclusion map $\mathbf{Ab} \rightarrow \mathbf{Grp}$ has a left-adjoint given by the abelianization: $G^{\text{ab}} = G/[G, G]$)
7. If you know topology, let $F : \mathbf{Top} \rightarrow \mathbf{Set}$ be the usual forgetful functor. This functor has a left and right adjoint: the left-adjoints gives the discrete topology to each set, while the right adjoint give the triiaval topology to each set.
8. Another famous topological examples is the Stone-Cech compactification theorem, where the inclusion functo $F : \mathbf{CHaus} \rightarrow \mathbf{Top}$ of compact hausdorff spaces into topological spaces has a left adjoint known mapping topological spaces to their Stone-Cech compactification.
9. An important adjoints that will appear in two major theorems later in the book^a comes when considering posets as categories (the objects are the elements and the morphisms are the relations, x has a morphism to y if and only if $x \leq y$). Then a pair of adjoint functors between two posets (interpreted as categories) is called a *galois connection*. If the functor is bijective and contravariant, then this is usually called a *galois correspondence*.

More adjoints shall apear later in each following chapter^b

^aGalois Correspondence and Nullstellensatz

^bThe hom-tensor adjoint and the Frobenius Reciprocity being two famous examples

Lemma 2.0.12: Continuous And Cocontinuous Functor

Let $F : C \rightarrow D$ be a right-adjoint functor. and $I \rightarrow C$ represent a diagram. Then:

$$F(\varprojlim A) \xrightarrow{\sim} \varprojlim (F \circ A)$$

if the limit exists. Similarly, if $G : C \rightarrow D$ is a right-adjoint functor

$$G(\varinjlim A) \xrightarrow{\sim} \varinjlim (G \circ A)$$

Analogously to calculus, when a function commutes with a limit, it is called *continuous*. Hence right-adjoint functors are continuous functors. Similarly, left-adjoint functors are *cocontinuous*.

Proof:

see ref:HERE

As a consequence, rightad-joint functors preserve producects and “preserve kernsl” when kernels make sense.

Noetherian Rings, Hilbert's Basis Theorem, and Localization

As with groups and modules, all R -algebras are quotient of a polynomial ring over many variables. If the polynomial ring is over finitely many variables. The fact that all algebras are quotient of polynomial rings will let the study of geometry and algebra.

Next, we recall some important definition and results that shall be central to the following section:

Definition 3.0.1: Noetherian Ring

Let R be a commutative ring. Then if R satisfies the ascending chain condition on Ideals, then R is Noetherian. In particular, if $\mathcal{I}$ is a collection of ideals ordered by inclusion $(I_1 \subseteq I_2 \subseteq \cdots I_k \cdots)$, then there exists an m such that for all $n \geq m$, $I_m = I_{m+1} = \cdots$.

Proposition 3.0.2: Extension of Noetherian Rings

Let R be a Noetherian ring. Then R/I is Noetherian for any ideal I . Conversely, if I and R/I is Noetherian, then R is Noetherian.

Proof:

This is proposition ref:HERE

As a consequence the image of any Noetherian ring is Noetherian.

Proposition 3.0.3: Equivalence of Noetherian Ring

Let R be a Noetherian ring. Then the following are equivalent

1. R is Noetherian
2. every nonempty collection of ideals of R has a maximal element under inclusion
3. every ideal of R is finitely generated

Proof:

this is proposition ref:HERE

Example 3.0.4: Exercise

Show that if R is a Noetherian ring and M is a finitely generated R -module, then M is Noetherian (Hint: remember the universal property of free-modules)

We will soon encounter rings that satisfy the descending chain condition (commonly abbreviated to D.C.C). Such rings are called *Artinian rings*. For the following result, let us upgrade ?? for modules:

Definition 3.0.5: Chain of Modules and Composition Series

Let M be a module. Then a *chain* of submodule of M is a sequence (M_i) of submodules st

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$$

where the length of the chain is the number of $\supsetneq$. A maximal chain is called a *composition series* of M (equivalently, each M_{i-1}/M_i is *simple*). If the chain is of infinite length, we shall say that M does not have a composition series.

Proposition 3.0.6: Finite Length, then Length is Invariant

Let M have a composition series of length n . Then every composition series of M has length n , and every chain in M can be extended to a composition series.

Proof:

Let $\ell(M)$ denote the least length of composition series of a module M . To get the desired result, let us prove a few results:

1. $N \subseteq M \Rightarrow \ell(N) \leq \ell(M)$. Let (M_i) be a composition series of M of minimum length, and take $N_i = N \cap M_i$. Then $N_{i-1}/N_i \subseteq M_{i-1}/M_i$, which as M_{i-1}/M_i is simple conclude the result
2. each $M_i \subsetneq M$ have length $\leq \ell(M)$. To see this, we may make a new chain with M_0 as the starting point:

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_k = 0$$

which by part 1 give us:

$$\ell(M) = \ell(M_0) \geq \ell(M_1) \geq \cdots \geq \ell(M_k) = 0$$

3. Take any composition series of M . Then if it has length k , it must be that $k \leq \ell(M)$ by part (2). Then by definition, $\ell(M)$, $k = \ell(M)$. Hence, all composition series have the same length.

Finally, consider any chain for M . If its length is $\ell(M)$, by (2) it must be a composition series, if its length is less than $\ell(M)$ it cannot be a composition series, and hence cannot be maximal. There does exist a lengthening by a new term, which as the number of possible terms that can be added is of finite length we may continue until we have a series of length $\ell(M)$, completing the proof.

3.1 Hilbert Basis Theorem

From ?? that if F is a field, then $F[x]$ is a Euclidean Domain. By corollary ??, we showed that $F[x_1, x_2, \dots]$ is a Unique Factorization Domain. However, of all the condition's on rings we have covered they are not necessarily stronger than Unique Factorization Domains (with the counter-example being example ?? where $\mathbb{Z}[x, y]$ is not a PID, and hence not a UFD). If we loosen the condition from the ideals be generated by a single element to being generated by *finitely many* elements (i.e. a Noetherian ring), then we in fact get that extending by an indeterminate x does not affect the finiteness condition on ideals.

Theorem 3.1.1: Hilbert's Basis Theorem (HBT)

Let R be a Noetherian ring. Then $R[x]$ is a Noetherian ring

If you know some linear algebra, the term “basis” is not like the term basis in linear algebra. It is because of an archaic use of the word basis when Hilbert formulated the theorem near the end of the 19th century, where it had a closer to “generation” than basis.

Proof:

Recall that if $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, then a_n is the leading coefficient of $p(x)$. This terminology will often be used in this proof.

Let $I \subseteq R[x]$ be an ideal. The trick is to somehow make I linked to an ideal of R (which is finitely generated since R is Noetherian). Let L be the set of all leading coefficients of the ideal I . Then L is an ideal. First, $0 \in I$, so $0 \in L$. Next, take $f, g \in I$ be two polynomials in I so that $f = ax^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ and $g = bx^m + b_{m-1}x^{m-1} + \cdots + b_1x + b_0$ with leading coefficients a and b and degrees n and m respectively, so that $a, b \in L$. Then since I is an ideal, $rf - g \in I$ which has leading coefficient $ra - b$, and so $ra - b \in L$. Now, since R is a Noetherian ring and L is an ideal of R , L is finitely generated, say $L = \langle a_1, a_2, \dots, a_n \rangle$. Then for each a_i , there is a polynomial f_i that has a_i as it's coefficient. Let's say e_i is the degree of f_i , so that the set $e_1, e_2, \dots, e_n$ represents the degree's of the polynomials representing the generators of L . Let $N = \max\{e_1, e_2, \dots, e_n\}$.

Next, we will construct a couple of smaller ideals. Let $d \in \{1, 2, 3, \dots, N-1\}$ (i.e. d is strictly smaller than the largest degree polynomial of L) and define $L_d \subseteq L$ to be the set of all leading coefficients of polynomials of I of degree d , together with 0. Then L_d is also an ideal of R as can be checked by a similar method as last proof. Since R is Noetherian, there exists a set of generators for L_d , say $L_d = \{b_{d,1}, b_{d,2}, \dots, b_{d,k_d}\}$. For each generator of L_d , there is a polynomial $f_{d,i} \in L_d$ that has the generator $b_{d,i}$ as it's leading coefficient.

Now, we will show that

$$I = \langle f_1, f_2, \dots, f_n, f_{1,1}, f_{1,2}, \dots, f_{1,k_1}, f_{2,1}, \dots, f_{N-1,k_{N-1}} \rangle$$

that is, I is generated by the polynomial f_i and the polynomials $f_{d,i}$. Since there are finitely many of them, I would be finitely generating completing the proof. To show this, let I' represent the right hand side. Then it is clear that $I' \subseteq I$, since every generator of I' is in I , and I' is the smallest set that contains those generators. So, let's prove $I \subseteq I'$, and take a $f \in I$.

If $\deg(f) \geq N$, then since $f \in I$, it's leading coefficient (say a) is in L : $a \in L$. Then since L is finitely generated, it is an R -linear combination of the generators of L : $r_1 a_1 + r_2 a_2 + \dots + r_n a_n$. From this, we can define a new polynomial $g_1 = r_1 x^{d-e_1} f_1 + r_2 x^{d-e_2} f_2 + \dots + r_n x^{d-e_n} f_n$ with the same leading coefficient a of f that is in I' . Then

$$f - g_1$$

is a polynomial in I of degree less than $\deg(f)$. If $\deg(f - g_1)$ is still greater than or equal to f , then repeat the same process, making the polynomial $(f - g_1) - g_2$. Continue on this way till we get a polynomial of degree less than N :

$$f - g_1 - g_2 - \dots - g_\ell = f - (g_1 + g_2 + \dots + g_\ell)$$

Let's label this above polynomial f' . Note that the sum of the g_i 's can be re-written as an R -linear combination of elements of I' since they all belong to I' :

$$f - (\alpha_1 f_1 + \alpha_2 f_2 + \dots + \alpha_n f_n)$$

Now, $\deg(f') < N$. Since $f' \in I'$, it's leading coefficient (say a) is in L_d : $a \in L_d$. Then since L_d is finitely generated, it is an R -linear combination of the generators of L_d : $a = r_1 b_{d,1} + r_2 b_{d,2} + \dots + r_{k_d} b_{d,k_d}$. From this, we can define a new polynomial $h_1 = r_1 f_{d,1} + r_2 f_{d,2} + \dots + r_{k_d} f_{d,k_d}$ with the same leading coefficient as f' that is in I' . Then

$$f' - h_1$$

Is of degree less than f' . Just as before, we can continue this procedure, which will terminate once the degree reaches 0:

$$f' - h_1 - h_2 - \dots - h_p = f' - (h_1 + h_2 + \dots + h_p)$$

Note that $\sum_i h_i$ can be decomposed into a sum of polynomials with leading coefficients from L_d :

$$f' - (\beta_{1,1} f_{1,1} + \beta_{1,2} f_{1,2} + \dots + \beta_{k,k_d} f_{k,k_d})$$

Since our algorithm terminated after we have subtracted the constant (that is, the leading term of L_0), we are left with:

$$f - (g_1 + g_2 + \dots + g_\ell + h_1 + h_2 + \dots + h_p) = 0$$

which implies $f = g_1 + g_2 + \cdots + g_\ell + h_1 + h_2 + \cdots + h_p$. Re-writing slightly, we get that

$$f = \alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_n f_n + \beta_{1,1} f_{1,1} + \beta_{1,2} f_{1,2} + \cdots + \beta_k k_d f_{k,k_d}$$

which shows that f is a combination of elements of I' and so $f \in I'$. Since f was arbitrary, we can conclude that $I \subseteq I'$.

Along with the quick assertion earlier that $I' \subseteq I$, that shows us that $I = I'$, showing that I was finitely generated. Since I was an arbitrary ideal, we are left with saying that every ideal of $R[x]$ is finitely generated, showing that $R[x]$ is also Noetherian, as we sought to show.

The result is not necessarily true if R is not Noetherian

In the exercises there will be the outlining of an alternative proof which will give a different insight (see exercise ref:HERE)

Corollary 3.1.2: HBT on multi-variable Polynomials

Let R be a Noetherian ring. Then $R[x_1, x_2, \dots, x_n]$ is a Noetherian ring

Proof:

Since R is a Noetherian Domain, by the Hilbert Basis Theorem, $R[x_1]$ is Noetherian. Since $R[x_1]$ is a Noetherian Domain, $(R[x_1])[x_2] = R[x_1, x_2]$ is Noetherian. Continuing Inductively, we see that $R[x_1, x_2, \dots, x_n]$ is Noetherian, as we sought to show

Corollary 3.1.3: f.g. K -algebra then Finitely Presented

Let R be a finitely generated k -algebra. Then R is *finitely presented*, meaning

$$R \cong \frac{k[x_1, \dots, x_n]}{(f_1, \dots, f_m)} = \frac{k[x_1, \dots, x_n]}{I}$$

where I is finitely generated.

Proof:

As R is finitely generated as a k algebra:

$$R \cong \frac{k[x_1, \dots, x_n]}{I}$$

for some finitely many variable $x_1, \dots, x_n$ and an ideal $I \subseteq R$. By corollary 3.1.2 R is Noetherian, hence I is *finitely generated*: $I = (f_1, \dots, f_m)$. But then R is finitely presented, completing the proof.

You can now return to example ?? to fully appreciate how there exist Atomic Domains which are not Noetherian Domains.

This theorem is very powerful. Since $R[x]$ is an ideal of itself, it is also finitely generated and the polynomials which generate it finitely can be characterized by their leading coefficients. In section 3.2, we will expand this theory further find a computational way of getting these polynomials.

Since a field is Noetherian (only ideal is (1) and (0), both clearly finite), Hilbert's Basis Theorem and induction immediately give:

Corollary 3.1.4: HBT on Polynomial Rings of Several Variables

Every ideal in the polynomial ring $F[x_1, x_2, \dots, x_n]$ with coefficients from a field F is finitely generated

Proof:
exercise

Recall that finitely generated R -algebras are quotient of polynomial rings R . By HBT, we have the following

Corollary 3.1.5: Finitely Generated Algebras Are Noetherian

Let A be a finitely generated R -algebra. Then if R is Noetherian, A is finitely generated if and only if it is Noetherian.

Proof:
exercise.

Polynomial rings being over a Noetherian ring being Noetherian is central to the study of algebras thanks to the universal property of Algebras

(A word on how Artinian rings are the opposite with D.C.C. We won't study them till section 7, since they sort of seem like a "oh wow, Noetherian rings can do so much, what about the opposite?" and then we realise that Artinian ring implies Noetherian ring, and it has so much structure that we can easily classify them all with no further fought, so we'll just leave that for later. All in all, the Noetherian condition turns out to be "more interesting")

3.1.6 Without Assuming Noetherian Let $R = k[x_1, \dots, x_n, \dots]$ and $A = R/(x_1, \dots, x_n, \dots)$. Then A is finitely generated, but the ideal is not finitely generated. In algebraic geometry, there is often the need to work with rings that look like quotients of polynomial rings (ex. polynomial rings over local rings). Thus, it common to work with *finitely presented* rings.

Exercise 3.1.1

1. Alternative HBT: (the one uses Diksons lemma)
2. Consider the chain $k[[x]] \subseteq k[[x^{1/2}]] \subseteq k[[x^{1/6}]] \subseteq \dots \subseteq k[[x^{1/n!}]]$. Taking the union of all of these forms an object called the *Puiseux series*. Is the Puiseux series a ring, and if so is the Puiseux series Noetherian?
3. Let S be a finitely generated R -algebra, where R is Noetherian. Prove that S is Noetherian.
4. We know that ideals of $k[x]$ are generated by 1 element if k is a field. By Hilbert's basis theorem, we know that every ideal of $k[x, y]$ is finitely generated. Is it true that ideals of

$k[x, y]$ is generated by at most 2 elements if k is a field?

5. Which of the following are Noetherian Rings:

1. $\mathbb{Z}[x, 1/x]$ as a subring of $\mathbb{Q}$
2. $R := \mathbb{R} + x\mathbb{C}[x]$
3. $R := k + xL[x]$ where $[L : k] < \infty$ is a finite field extension
4. $R := k + xk[x, y]$ as a subring of $k[x, y]$, k a field.

6. Let M be an R -module over a Noetherian ring R . Show that there exists a $x \in M$ such that $\text{Ann}_R(x)$ is a prime ideal (large hint: consider the set $\mathcal{S} = \{\text{Ann}_A(x) \mid x \in M\}$ of ideals, and use the fact that R is Noetherian to conclude $\mathcal{S}$ has a maximal element. Is this element prime?)

3.2 Gröbner Basis

here?

3.3 Localization

Localization shall be the generalization of section ?? to commutative rings and modules in general (including ones with zero-divisors). With the introduction of the tensor product, localization shall be shown to have the important property of producing *flat module*, which shall have important geometric consequences in algebraic geometry. From this, we shall define the notion of local properties and the famous Nakayama lemma which shall allow many proofs to be done “locally”. Concluding this section, we shall wrap up the correspondence between primes of a ring R and primes in its integral extension.

The term “localization” comes from algebraic geometry, where this process of localizing mirrors the process of “localizing” for manifolds. If you have done some differential geometry, recall that the tangent space at a point p can be defined as the set of all germs at a point p whose value is 0, and then taking its quotient with its square: $\mathfrak{m}_p/(\mathfrak{m}_p)^2$. Localizing shall correspond to this process, where “inverting” elements around a point is the same as a function on a manifold being invertible on a small enough local neighborhood where $f(p) \neq 0$. The reader may question this correspondence, for in differential geometry we are working with smooth functions over a [smooth] manifold, while in the algebraic case we simply have a [commutative] ring. However, the correspondence is more equivalent than it may at first seem. It shall be particularly clear by the end of chapter ??, where it shall be shown that there is a way of thinking of all commutative rings as a set of “functions” on a special geometric space (the geometric space is called an [affine] *scheme*), hence there is a direct parallel between smooth functions on the geometric object of a manifold and commutative rings where each element is interpreted as a function on a scheme.

3.3.1 Localization of Rings

This is a generalization of the constructing of ring of fraction from ?? by allowing the set that's closed under multiplication D to have zero divisors. This will imply that R need not be embedded into

$D^{-1}R$, but the universal property is still satisfied. Furthermore, we will have to solve the problem of elements that “should” be zero being equal to nonzero elements. If $a \in R$ and $d \in D$ where $as = 0$, then:

$$a = \frac{a}{1} = \frac{as}{s} = \frac{0}{s} = 0$$

this issue will be addressed in the proof:

Theorem 3.3.1: Universal Property of Localization

Let R be a commutative ring and D be a multiplicatively closed subset containing 1. Then there exists a ring $D^{-1}R$ such that the following diagram commutes

$$\begin{array}{ccc} D^{-1}R & & \\ \uparrow \iota & \searrow \Psi & \\ R & \xrightarrow{\psi} & S \end{array}$$

with the property that $\psi(D)$ is a subset of the units of S

Proof:

A lot of this proof is similar to ?? (Ring of Fractions) and so it will be quickly presented pointing out where the generalization of D having zero divisors makes a difference.

Let $R \times D$ have the equivalence relation

$$(a, b) \sim (c, d) \Leftrightarrow x(ad - bc) = 0 \text{ for some } x \in D$$

Which is essentially the same as in theorem ?? . The difference between the proof in the case where D does not have zero divisors' is with the fact that the terms are scaled by some $x \in D$. This is to make sure elements that are zero-divisors will belong to the $[0]$ (as we'll show in the proceeding corollary). Usually, the equivalence class $[(r, d)]$ is denoted $\frac{r}{s}$ or r/s so that

$$\frac{a}{b} = \frac{c}{d} \text{ if and only if } x(ad - bc) = 0 \text{ for some } x \in D$$

Define

$$\frac{r}{d} + \frac{s}{e} := \frac{re + sd}{de} \quad \frac{r}{d} \frac{s}{e} := \frac{rs}{de}$$

Then these two operations form a well-defined commutative ring $(R \times D)/\sim$; this ring is denoted $D^{-1}R$. Note that $\frac{d}{1}$ is always a unit in $D^{-1}R$ (even when it is a zero-divisor, since it will be shown the ring will be a zero-ring)

Define $\iota : R \rightarrow D^{-1}R$ by $\iota(r) = \frac{r}{1}$. This map defines a natural ring homomorphism. Using this map, if the map $\varphi : R \rightarrow S$ maps D to units, we can define the map $\Psi : D^{-1}R \rightarrow S$ by defining:

$$\Psi\left(\frac{r}{d}\right) = \varphi(r)\varphi(d)^{-1}$$

This map is well-defined: Notice if $\frac{r}{d} = \frac{s}{e}$ then by definition of the equivalence relation, for some $x \in D$, $x(re - sd) = 0$. Thus $\varphi(x(re - sd)) = \varphi(x)(\varphi(re) - \varphi(sd)) = 0$. Since x is a unit in S and S is a commutative ring, s is not a zero-divisor, and so $\varphi(re) - \varphi(sd) = 0$. Moving values around, we get

$$\varphi(r)\varphi(d)^{-1} = \varphi(s)\varphi(e)^{-1}$$

Thus, Ψ is a well-defined ring homomorphism. Furthermore, by construction, $\Psi \circ \iota = \varphi$

Finally, Ψ is unique. If there was another Ψ' such that $\Psi' \circ \iota = \varphi$, then:

$$\varphi(r) = \Psi'(\iota(r)) = \Psi'\left(\frac{r}{1}\right)$$

Thus Ψ' maps $\frac{r}{1}$ in the same way as Ψ . For the $\frac{1}{d}$ terms, notice that

$$\frac{1}{d} = \left(\frac{d}{1}\right)^{-1}$$

and so $\varphi(d) = \Psi'\left(\frac{d}{1}\right)$, and so, by properties of ring homomorphisms

$$\Psi'\left(\frac{d}{1}\right)^{-1} = \Psi'\left(\frac{1}{d}\right)$$

showing that $\Psi' = \Psi$, fulfilling the last condition needed to prove for the universal property

As before, $D^{-1}R$ is still called the ring of fractions. However, there is a more common name with a geometric intuition:

Definition 3.3.2: Localization of R at D

Let R be a commutative ring and D be a multiplicatively closed subset of R . Then the ring $D^{-1}R$ produced in the previous theorem is the *ring of fractions of R with respect to D* , or the *localization of R at D* . This ring is also sometimes denoted $R[D^{-1}]$.

If $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$ for some $f \in R$, we usually denote $D^{-1}R = R_f$.

If D is not a multiplicatively closed set, we define $R[D^{-1}] := R[\overline{D}^{-1}]$ where $\overline{D}^{-1}$ is the multiplicative closure of D .

The fact that D can contain zero divisors means that ι can now have non-trivial kernel:

Corollary 3.3.3: Kernel of ι

Let ι be the natural inclusion of R into $D^{-1}R$ for some appropriate D . Then

1. $\ker(\iota) = \{r \in R \mid xr = 0 \text{ for some } x \in D\}$. In particular, ι is an injection if and only if D doesn't contain 0 or zero-divisors
2. $D^{-1}R = 0$ if and only if D contains no nilpotent elements or 0 (since D is multiplicatively closed)

Proof:

1. If $\iota(r) = \frac{r}{1} = 0$, then we have that $(r, 1) \sim (0, 1)$ and so $x(1r - 0) = xr = 0$ for some $x \in D$. Thus, ι is an injection if and only if D does not contain 0 or zero-divisors
2. $D^{-1}R = 0$ if and only if $r/1 = 0/1$ for all r i.e. $(r, 1) \sim (0, 1)$ if and only if $(1, 1) \sim (0, 1)$ if and only if $x \cdot 1 = 0$. If $x = 0$, we're done, and if not then as D is multiplicatively closed

then $x^n \in D$ for each $n \in \mathbb{N}$ and so if $x^n = 0$ then x is nilpotent and $x^n \cdot 1 = 0$, completing the proof.

This let's us get a sense of how adding zero-divisors to D changes the resulting localization of R at D .

Example 3.3.4: Localization

All examples in section ?? are valid examples, including the trivial one of any field $\mathbb{F}$, $\mathbb{Z}$ to $\mathbb{Q}$ with $D = \mathbb{Z} - \{0\}$, and $\mathbb{F}[x]$ to $\mathbb{F}(x)$ with $D = \mathbb{F}[x] - \{0\}$. We shall give some more interesting examples here (all rings are commutative):

1. If R is an integral domain, then we can always take $D = R - \{0\}$. The ring $D^{-1}R$ is then always a field, and is called the *field of fractions of R* , and is sometimes denoted Q . Then for any smaller multiplicatively closed subset E , $E^{-1}R$ is a subring of Q with elements $r/e \in Q$ with $r \in R$ and $e \in E$
2. Let R be a ring and $f \in R$. Define $D = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$. Then we'll denote $D^{-1}R = R_f$. Then R_f is a ring in which the element f has become a unit. In a sense, it is now possible to solve the question

$$fx - 1 = 0$$

In fact, this intuition can be made more rigorous showing that

$$R_f \cong \frac{R[x]}{fx - 1}$$

This is similar to how we worked with field extensions, and in fact is an important link to connect the intuition of localization to a geometric intuition. Keep this example in the back of your mind for now.

3. Consider $R = k[x, y]/(xy)$, take $\bar{x} \in R$, and consider $R_{\bar{x}}$. Then the the map $\iota : R \rightarrow R_{\bar{x}}$ is no longer injective as:

$$\bar{y} = \frac{\bar{x}\bar{y}}{\bar{x}} = \frac{\bar{0}}{\bar{x}} = \bar{0}$$

4. This example is called *localizing at a prime*. Let R be a ring and P a prime ideal of R . Then the set $D = R - P$ is multiplicatively closed. To see that, let $x, y \in D$. If it were the case that $xy \in P$, then either $x \in P$ or $y \in P$. But then one of them wouldn't be in D – a contradiction. Thus, D is indeed multiplicatively closed. The localization $D^{-1}R$ is called the *localization of R at P* and is commonly denoted as R_P . In this ring, all elements not in P become units. As a simple example, take $R = \mathbb{Z}$ and $P = (p)$ for some prime p . Then by the nice divisibility properties of integers, the set $\mathbb{Z}_{(p)}$ can with a little work be shown to be:

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} \mid p \nmid b \right\}$$

that is, it is all elements of $\mathbb{Q}$ for which the denominator is not divisible by p . Can you prove that $\mathbb{Z}_{(p)}$ has only one maximal ideal (and hence, is a local ring)? Is this true of all localization at primes?

It is tempting to say that if $R_{(p)}$ is the localization of a ring R at the prime p and $a/b \in R_{(p)}$ if $b \notin (p)$, however this is not necessarily the case as we are dealing with an equivalence class (exercise [ref:HERE](#))! The reader may then ask if we can take a better representative (like the representative a/b in $\mathbb{Z}$ such that $\gcd(a, b) = 1$), however R is not necessarily a UFD and so this may not be well-defined (exercise [ref:HERE](#)[ref:HERE](#)).

5. The following examples is a way to “localise” functions. Let k be a field and V be any set. Define R to be functions from V to k containing the constant functions ($F \subseteq \text{Hom}_{\text{Set}}^+(V, k)$) with $+$ and $\cdot$ defined by component-wise addition and multiplication on the image: $R = (F, +, \cdot)$. For example, take all continuous functions from $[0, 1]$ to $\mathbb{R}$. For any $a \in V$, let $M_a \subseteq R$ be the set of all functions that vanish at a : $f(a) = 0$. Then M_a is the kernel of the evaluation map from R to k :

$$\text{ev}_a : R \rightarrow k \quad f \mapsto f(a) \quad \ker(\text{ev}_a) = M_a$$

Since R contains constant functions, ev_a is surjective and so $k \cong R/M_a$. By ??, M_a is a maximal ideal, and hence a prime ideal. Thus, we can define the localization of R at M_a to be:

$$R_{M_a} = \left\{ \frac{f}{g} \mid f, g \in R, g(a) \neq 0 \right\}$$

where ev_a is extended as:

$$(f/g)(a) = f(a)/g(a)$$

This then makes R_{M_a} a way of defining rational functions.

6. Let $R = \mathbb{C}[x]$, and take $p = (x - a)$. Since $\mathbb{C}[x]$ is a Principal Ideal Domain and $(x - a)$ is irreducible, $(x - a)$ is prime, and so we can define

$$\mathbb{C}[x]_{(x-a)} = \{p(x)/q(x) \mid x - a \nmid q(x)\}$$

which can be thought of all rational functions which do not vanish at a .

Localization has many nice geometric interpretations. As we will see in section ??, all algebraic varieties (i.e. irreducible algebraic sets) are related to the prime ideals. Hence, understanding how ideals of R correspond to ideals of $D^{-1}R$ will bring us to understanding a geometric connection.

For notational simplicity, let $D^{-1}R\iota(I) = {}^{\iota}I$ for some ideal I (note that this is *not* $\iota(I)$). Let $\pi : D^{-1}R \rightarrow R$ be the function mapping $r/d \mapsto r$ so that for some ideal $I \subseteq D^{-1}R$, $R\pi(I) = \iota^{-1}(I) = {}^{\pi}I$. The ideal ${}^{\iota}I$ is sometimes called the *extension* of I and ${}^{\pi}I$ the *contraction* of I . Since every element of ${}^{\iota}I$ can be written of the form a/d where $a \in I$ and $d \in D$, the notation $D^{-1}I$ is also frequently used.

Proposition 3.3.5: Ideals of Localization

Let R be a ring and $D^{-1}R$ be the localization of R at D for appropriate D . Then:

1. For every ideal $J \subseteq D^{-1}R$,

$$J = J^{ce}$$

As a consequence, every ideal of $D^{-1}R$ is an extension of an ideal in R , and every ideal of $D^{-1}R$ contracts to a distinct ideal in R

2. If $I \subseteq R$ is an ideal of R , then:

$$(\lim I)^e = \{r \in R \mid dr \in I \text{ for some } d \in D\}$$

As a consequence, $I^e = D^{-1}R$ if and only if $I \cap D \neq \emptyset$

3. There is a bijection between the set of prime ideals with $P \cap D = \emptyset$ and the set of prime ideals of $D^{-1}R$:

$$\left\{ \begin{array}{c} \text{prime ideals of } R \text{ with } \\ P \cap D = \emptyset \end{array} \right\} \xrightleftharpoons[(-)^c]{(-)^e} \{\text{prime ideals of } D^{-1}R\}$$

4. If R is Noetherian, then $D^{-1}R$ is Noetherian. Similarly if R is Artinian

Proof:

1. Notice that by construction, ${}^e(\pi J) \subseteq J$. For the other direction, notice that for any $a/d \in J$ it is always the case that $a/1 = d(a/d) \in J$, that is there exists a a/d such that $d(a/d) = a/1$. Since $\iota(\pi(a/d)) = a/1$. and so by definition, of ${}^e(\pi J)$, $(1/d)(a/1) = a/d \in {}^e(\pi J)$ (since it closed by multiplication of elements from $D^{-1}R$).
2. Let $I' = \{r \in R \mid dr \in I \text{ for some } d \in D\}$ be the set in the question. We'll show $I' = {}^\pi(\lim I)$. For the $\subseteq$ direction, take $r \in I'$. Then by definition there exists a $d \in D$ such that $dr = a \in I$. Then $r/1 = a/d \in \lim I$, so contracting we get $r \in {}^\pi(\lim I)$. For the $\supseteq$ direction, take $r \in {}^\pi(\lim I)$. Then $r/1 = a/d \in \lim I$ for some $a \in I$ and $d \in D$. Then by the definition of the equivalence class, we have $x(dr - a) = 0$ for some $x \in D$, and hence $xdr = xa$ where $xa \in I$. Since $xd \in D$, $r \in I'$. The rest of the assertions of (2) follow quickly.
3. To map prime ideals from $D^{-1}R$ to R , recall that pre-images of prime ideals is prime (see exercise ref:HERE in section ref:HERE). Thus, by construction, the contraction must map prime ideals of $D^{-1}R$ to prime ideals that are disjoint from D .

For the converse, let I be an ideal in R that is disjoint from D , and consider $Q = {}^e I$. We'll show Q is prime. Suppose $(a/d_1)(b/d_2) = (ab/d_1d_2) \in Q$. Then $ab/d_1d_2 = c/d$ for some $c \in P$ and $d \in D$. Then by definition of the equivalent relation, $x(dab - d_1d_2c) = 0$ for some $x \in D$. Since $xdab - xd_1d_2c = 0 \in P$ and P is closed under subtraction and left-multiplication of elements of R , it must be that $xdab = (xd)(ab) \in P$. Since P is disjoint from D , it must be that $ab \in P$. Since P is prime, either a or b is an element of P . Thus, either a/d_1 or b/d_2 is an element of Q , showing that Q is a prime ideal. Thus,

extending prime ideals that are disjoint from D map to prime ideals in $D^{-1}R$.

Finally, by (2), since P is disjoint from D we have $P = \pi({}^e P)$, showing that extending and contracting are inverses of each other, finishing the proof for (3).

4. Given some ascending chain of ideals in $D^{-1}R$, they contract to ascending chain of ideals in R . Since R is Noetherian, they stabilize. Thus, the ideals that get contracted must also stabilize. Similarly for Artinian Rings.

The ideal in part (2) of proposition 3.3.5 is given a name:

Definition 3.3.6: Saturated Ideal

Let R be a commutative ring, $I \subseteq R$ is an ideal and $D^{-1}R$ is the localization of R at D for appropriate D . Then *saturation* of the ideal I with respect to D is $\pi(\lim I)$. If $I = \pi(\lim I)$, then I is said to be *saturated* or *D-saturated* with respect to D .

(Re-write the previous results in terms of $\text{Spec } R$)

and show that for localization at a prime, we have the following image:

If we instead have R_f to be the localization at f (so $D = \{f^n \mid n \in \mathbb{Z}\}$), then in fact there is a bijection between $\text{Spec } R \cong_{\text{Set}} \text{Spec } R_f \oplus \text{Spec } R/(f)$. Later on, we will see that we can upgrade this bijection into a homeomorphism

Proposition 3.3.7: Localization Property Preservation

Let R be a ring and let $D^{-1}R$ be the localization of R . Then the following properties of R are preserved in $D^{-1}R$:

1. no zero divisors
2. Noetherian
3. Integrally closed

Proof:
exercise

Exercise 3.3.1

1. Let R be a ring, f be an element of R , and R_f be the localization of R at $\{f^n \mid n \in \mathbb{Z}_{\geq 0}\}$. Then

1.

$$R_f \cong \frac{R[x]}{fx - 1}$$

2. $R_f \cong R_{f^n}$ for $n \geq 1$

3. If f is nilpotent, then $R_f = 0$

2. Let R be a normal domain and S a multiplicatively closed set. Then $S^{-1}R$ is a normal domain

3.3.2 Localization of Modules

Since every ring is a module over itself, we can see how localization works for this special case of modules. We extend the concept of localization now to more general modules. The construction is essentially the same thing: if M is an R -module, and $D \subseteq R$ is multiplicatively closed, define the same equivalence class on $M \times D$ and the operation's on $(M \times D)/\sim$ as

$$\frac{m}{d} + \frac{m'}{d'} = \frac{d'm + dm'}{dd'} \quad \text{and} \quad r \cdot \frac{m}{d} = \frac{r \cdot m}{d}$$

Then this is indeed a module with the proof proceeding essentially the same as theorem 3.3.1, and so is left as a sanity check. Just like $R[D^{-1}]$ is the localization of R at D , we will denote $M[D^{-1}]$ to be the localization of M at D . Thus, an R -module M is extended to an $D^{-1}R$ -module $M[D^{-1}]$.

Proposition 3.3.8: Zeros of $M[D^{-1}]$

Let R be a ring, $D \subseteq R$ be a multiplicatively closed subset, and M be a module over R . Then m goes to 0 in $M[D^{-1}]$ if and only if m is annihilated by an element $d \in D$. If M is finitely generated, then $M[D^{-1}] = 0$ if and only if M is annihilated by an element in D .

If R lies in a field K and M is an R -module that is contained in a k -vector space, then $\iota : M \rightarrow D^{-1}M$ is injective iff $M \xrightarrow{s} M$ is injective for all $s \in D$.

Proof:

Exercise

Localization modules works well with homomorphisms: let $\varphi : M \rightarrow N$ be an R -module homomorphism. Then by the universal property, we see that this is extended to an $D^{-1}R$ -module homomorphism $D^{-1}\varphi : M[D^{-1}] \rightarrow N[D^{-1}]$ by mapping:

$$D^{-1}\varphi\left(\frac{m}{d}\right) = \frac{\varphi(m)}{d}$$

Furthermore, $D^{-1}(\varphi \circ \psi) = (D^{-1}\varphi) \circ (D^{-1}\psi)$, i.e. $D^{-1}(-)$ is a functor from Mod_R to $\text{Mod}_{D^{-1}R}$. We may ask for certain properties of this functor, most basic perhaps would be if $D^{-1}(-)$ is an exact functor, that it preserves kernels and cokernels. This is the case:

Proposition 3.3.9: Localization Is Exact

Let M_1, M_2, M_3 be R -modules such that $0 \rightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \rightarrow 0$ is a short exact sequence. Then

$$0 \rightarrow M_1[D^{-1}] \xrightarrow{D^{-1}f} M_2[D^{-1}] \xrightarrow{D^{-1}g} M_3[D^{-1}] \rightarrow 0$$

is a short exact sequence, that is $D^{-1}(_)$ is an exact functor.

Proof:

since $g \circ f = 0$, $D^{-1}g \circ D^{-1}f = D^{-1}(0) = 0$, and so $\text{im}(D^{-1}f) \subseteq \ker(D^{-1}g)$. For the reverse inclusion, let $m/d \in \ker(D^{-1}g)$ so that $g(m/d) = \frac{g(m)}{d} = 0$ in $M_3[D^{-1}]$. Thus, there exists an s such that $sg(m) = 0$ in M_3 . Since g is an R -module homomorphism, $sg(m) = g(sm)$, and so $sm \in \ker(g) = \text{im}(f)$, implying for some $m' \in M_3$, $f(m') = sm$. Putting it all together, we get that in $M_2[D^{-1}]$:

$$\frac{m}{d} = \frac{f(m')}{ds} = (D^{-1}f) \left(\frac{m'}{ds} \right) \in \text{im}(D^{-1}f)$$

showing that $\ker(D^{-1}g) \subseteq \text{im}(D^{-1}f)$, which along with the fact that $\text{im}(D^{-1}f) \subseteq \ker(D^{-1}g)$ shows that

$$\ker(D^{-1}g) = \text{im}(D^{-1}f)$$

completing the proof.

Notice that as we are defining a $D^{-1}R$ -module, we are extending scalars. If we use section ??, then we can use the following observation to arrive at this result more quickly:

Proposition 3.3.10: Characterization of Localization

Let M be an R -module and $M[D^{-1}]$ a localization. Then:

$$M[D^{-1}] \cong_R D^{-1}R \otimes_R M$$

Proof:

Define the map:

$$(r/d) \times m \mapsto \frac{r \cdot m}{d}$$

then it is easy to see that this map is R -bilinear, and hence by the universal property of the tensor-product, there exists an R -module homomorphism:

$$f : D^{-1}R \otimes M \rightarrow M[D^{-1}]$$

It is clear that f is surjective and is uniquely defined with respect to the universal property. It remains to check that f is injective. First, show that for any element in $D^{-1}R \otimes M$, it can be written as:

$$\frac{1}{d} \otimes m$$

for some $d \in D$ and $m \in M$. Let's say $f(1/d \otimes m) = m/d = 0$. Then $sm = 0$ for some $s \in D$. Then:

$$\frac{1}{d} \otimes m = \frac{s}{ds} \otimes m = \frac{1}{ds} \otimes sm = \frac{1}{ds} \otimes 0 = 0$$

showing that f is injective, and hence an R -module isomorphism, completing the proof.

As a direct consequence, $D^{-1}R$ is a *flat* R -module.

Corollary 3.3.11: Localization Commuting With Tensor

Let M and N be R -modules, $D \subseteq R$ a semi-group. Then there exists a unique isomorphism:

$$\varphi : M[D^{-1}] \otimes_{D^{-1}R} N[D^{-1}] \rightarrow (M \otimes_A N)[D^{-1}] \quad (m/d) \otimes (n/s) \mapsto (m \otimes n)/ds$$

In particular, if p is any prime ideal:

$$M_p \otimes_{A_p} N_p \cong (M \otimes_A N)_p$$

Proof:

Use proposition 3.3.10 and the appropriate isomorphism theorem for modules.

(Some more cool results from Micheal's p. 43)

3.3.3 Local Properties

Localization allows us to diminish our problem to a ring much closer to a field, in particular if we want to study a family of prime ideals contained in some prime ideal p . In fact, there are many results about rings that can be deduced by studying rings *locally*, namely by studying rings at each prime ideal¹. This motivates the following concept:

Definition 3.3.12: Local Properties

Let P be a property of a ring (ex. commutative, free, etc.). Then a property P of a ring R (resp. of an R -module M) is called a *local property* if it respects the following:

R (resp. M) has property $P \Leftrightarrow R_p$ (resp. M_p) has property P for each prime ideal $p \subseteq R$

In otherwords, a property is local if we can move back and forth between studying any ring (resp. module) localized at a prime. Here are some examples of local properties:

Proposition 3.3.13: Zero Module Is A Local Property

Let M be an R -module. Then the following are equivalent:

1. $M = 0$
2. $M_p = 0$ or all prime ideals $p \subseteq R$
3. $M_m = 0$ for all maximal ideal $m \subseteq R$

Proof:

Certainly $(1) \Rightarrow (2) \Rightarrow (3)$. For $(3) \Rightarrow (1)$, let $0 \neq x \in M$ be a nonzero element of an R -module M with $0 \neq 1$. Let $I = \text{Ann}(x)$. Then I is contained in some maximal ideal m : $I \subseteq m$. Consider M_m . Since $M_m = 0$, $x/1 = 0$, implying $rx = 0$ for some $r \in R - m$. However $\text{Ann}(x) \subseteq m$, meaning so such element can exist, a contradiction, showing that $M = 0$ as we sought to show.

¹This observation is at the basis of interpreting rings as geometric objects

Another property that only needs to be verified locally is the injectivity/surjectivity of a function:

Proposition 3.3.14: Injectivity/Surjectivity/Isomorphism Is A Local Property

Let $\varphi : M \rightarrow N$ be an R -module homomorphism. Then the following are equivalent:

1. φ is an injection/surjection/isomorphism
2. $\varphi_p : M_p \rightarrow N_p$ is injection/surjection/isomorphism for all prime ideals $p \subseteq R$
3. $\varphi_m : M_m \rightarrow N_m$ is injection/surjection/isomorphism for all maximal ideals $m \subseteq R$

Proof:

The proof shall be doen for injectivity, surjectivitiy follows easily and isomorphism is combining the two.

- (1) $\Rightarrow$ (2) Since $0 \rightarrow M \xrightarrow{\varphi} N$ is exact, by proposition 3.3.9, $0 \rightarrow M_p \xrightarrow{\varphi_p} N_p$ is exact, hence φ_p is injective for any prime ideal $p \subseteq R$.
- (2) $\Rightarrow$ (3) Every maximal idael is prime
- (3) $\Rightarrow$ (1) Let $K = \ker(\varphi)$. Then the sequence $0 \rightarrow K \rightarrow M \rightarrow N$ is exact, and hence proposition 3.3.9 $0 \rightarrow K_m \rightarrow M_m \rightarrow N_m$ is exact, therefore $K'_m \cong \ker(\varphi_m) = 0$ since φ_m is injective. Thus, by proposition 3.3.13 φ is injective.

to show surjectivity, simply reverse the arrows.

Corollary 3.3.15: Exactness Is A Local Property

Let A, B, C be R -modules and let $1 \rightarrow A_m \rightarrow B_m \rightarrow C_m$ for each maximal ideal $m \subseteq R$. Then there exists maps such that

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

is exact

Proof:

take advantage of the fact that injectivity/surjectivity are local properties

A final example that will be useful is that faltness is a local property:

Proposition 3.3.16: Flatness Is A Local Property

Let M be an R -module. Then the following are equivalent:

1. M is a flat R -module
2. M_p is a flat R_p -module for each prime ideal $p \subseteq R$
3. M_m is a flat R_m -module for each maximal ideal $m \subseteq R$

Proof:

(1) $\Rightarrow$ (2) comes from pervious lemmas and (2) $\Rightarrow$ (3) is immediate. For (3) $\Rightarrow$ (1), let $N \rightarrow P$ be any R -module homomorphism and $m \subseteq R$ any maximal idea. Then:

$$\begin{aligned}
 N \rightarrow P \text{ is injective} &\Rightarrow N_m \rightarrow P_m \text{ is injective} && \text{proposition 3.3.14} \\
 &\Rightarrow N_m \otimes_{R_m} M_m \rightarrow P_m \otimes_{R_m} M_m \text{ is injective} && \text{flatness property} \\
 &\Rightarrow (N \otimes_R M)_m \rightarrow (P \otimes_R M)_m \text{ is injective} && \text{corollary 3.3.11} \\
 &\Rightarrow N \otimes_R M \rightarrow P \otimes_R M \text{ is injective} && \text{proposition 3.3.14}
 \end{aligned}$$

Hence, M is flat.

Proposition 3.3.17: Integral Closure Is A Local Property

Let R be an integral domain. Then the following are equivalent:

1. R is integrally closed
2. R_p is integrally closed for each prime ideal $p \subseteq R$
3. R_m is integrally closed for each maximal ideal $m \subseteq R$

Proof:

exercise.

The following is a way of proving results locally:

Proposition 3.3.18: Local Properties On Modules

Let R be a subring of a field k , and let M be an R -module contained in a k -vector space V (equiv. the map $M \rightarrow M \otimes_R k$ is injective). Then:

$$M = \bigcap_{\mathfrak{m}} M_{\mathfrak{m}} = \bigcap_{\mathfrak{p}} M_{\mathfrak{p}}$$

Proof:

Certainly $M \subseteq \bigcap_{\mathfrak{m}} M_{\mathfrak{m}}$. For the other direciton, take $x \in \bigcap_{\mathfrak{m}} M_{\mathfrak{m}} - M$. For each maximal ideal $\mathfrak{m}$, $x = m/s$ for some $m \in M$ and $s \in R - \mathfrak{m}$. Hence $sx \in M$. Define the ideal

$$I_a = \{a \in R \mid ax \in M\}$$

Then $as \in I_a$. But by construction $s \notin \mathfrak{m}$, hence $I_a \not\subseteq \mathfrak{m}$. But then it must be that $I_a = R$, hence $x = 1 \cdot x \in M$, the giving the result

Extending to prime ideals is left as an exercise.

Corollary 3.3.19: Local Properties Of Ideals

Let R be an integral domain. Then every ideal $I \subseteq R$ is equal to the intersection of its localizations at the maximal ideals of R , and also to the intersections of its localization at the prime ideals of R

Hence, to show a property of an ideal I in an integral domain R , it suffices to show that the property holds for all its localization $I_{\mathfrak{p}}$ and it is preserved under intersections.

Exercise 3.3.2

1. Show that M is projective if and only if each M_p is free.

3.3.4 Local Rings

Recall that the localization of $\mathbb{Z}$ at any prime ideal (p) , $\mathbb{Z}_{(p)}$, has a unique maximal ideal. It was left as an exercise to see whether any localization at a prime, R_p left R with only a single maximal ideal. This is indeed the case, meaning there is a unique field associated to R_p , letting us study R in terms of the collection of fields induced by R_p . Thus, for any integral domain R , we can always produce a collection of rings that only have a single maximal ideal $\mathfrak{m}_p$. Once we interpret a ring as a geometric object $\text{Spec}(R)$, we shall see that the rings $R_{\mathfrak{p}}$ representing the local properties of $\text{Spec}(R)$, not unlike how we have stalks at a point in Differential Geometry. Due to this, it is worth-while studying rings with a unique maximal ideal:

Definition 3.3.20: Local Ring

Let R be a ring (not necessarily commutative). Then if R has a unique two-sided maximal ideal, then R is said to be a *local ring*.

As a consequence, all simple rings (including fields and $M_k(n)$ for $n \geq 2$) are trivially local rings.

Proposition 3.3.21: Local Ring Relation to left/right Maximal Ideal

Let R be a ring (not necessarily commutative). Then the following are equivalent:

1. R is a local ring
2. R has a unique maximal left-ideal
3. R has a unique maximal right-ideal

Proof:
exercise

In the commutative case, this translates to R simply having a unique maximal ideal. This is particularly enlightening: if x is not in M , then x must be a unit, or else $(x) \subseteq M$.

Example 3.3.22: Local Rings

1. Every field is trivially a local ring.
2. $\mathbb{Z}/p^n\mathbb{Z}$ is a local ring with $n > 1$ with maximal ideal $\bar{p}$
3. if $f \in k[x]$ is irreducible then $k[x]/(f(x)^n)$ is a local ring with maximal ideal $\overline{(f(x))}$
4. (geometric example) Let M be a smooth manifold (real or complex). Then define $\mathcal{O}_{M,x}$ to be the germ of smooth functions at x , that is, two smooth functions $f, g \in C^\infty(M)$ are in the same equivalence relation $f \sim g$ (i.e. $[f] = [g]$ are the same germ) if there exists some open set $U \subseteq M$ containing x such that $f|_U = g|_U$. Clearly, $\mathcal{O}_{M,x}$ is a ring (using the ring operation of the codomain $\mathbb{R}$ or $\mathbb{C}$). The maximal ideal of $\mathcal{O}_{M,x}$ are those functions that are 0 at x

In fact, this last example will have a direct parallel in scheme theory, see [NateAlgGeo].

Proposition 3.3.23: Local Ring Equivalences

Let R be a local ring. Then the following are equivalent:

1. R is local ring.
2. $1 \neq 0$, and the sum of any two non-units is a non-unit
3. $1 \neq 0$, and if $x \in R$ then either x is a unit, or $1 - x$ is a unit
4. If a finite sum is a unit, then one term has to be a unit

Notice too that the final condition can include the “vacuously empty” sum, which sums to 0, and since 0 is not a unit, $0 \neq 1$ (TBD).

Proof:

here

Using localization and quotienting, we have a large amount of control over the prime ideals we want to work with: given a (commutative) ring R , if we take R_p , we eliminates all ideals except those contained in p , while R/p eliminates except those contained not contained in p (i.e. containing p). This can be visualized like so:



Hence, R_p/q^e (resp. $(R/p)_{\bar{q}}$) will only keep all the ideals between p and q , and if $p = q$, then the result is a field, known as the *residue field*:

Definition 3.3.24: Residue Field

Let R be a commutative local ring with maximal ideal m . Then R/m is called the *residue field*.

The quotient $R/\mathfrak{p}$ and the ring $R_{\mathfrak{p}}$ are canonically related: the ring $R_{\mathfrak{p}}$ can be thought of as the localization of $R/\mathfrak{p}$:

Proposition 3.3.25: Localization and Quotient Commute

Let $\mathfrak{p} \subseteq R$ be prime ideal of a ring R , and let $\mathfrak{m}_{\mathfrak{p}}$ be the unique maximal ideal of $R_{\mathfrak{p}}$. Then localizing at $\mathfrak{p}$ and then taking the field of fractions is isomorphic to taking quotient at $\mathfrak{p}$ and then localizing (at $\overline{f\mathfrak{p}} = \overline{0}$). Then there is a canonical embedding:

$$R/\mathfrak{p} \hookrightarrow R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$$

that maps $R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$ into the field of fractions of $R/\mathfrak{p}$

This may be visualized as the following diagram:

$$\begin{array}{ccccc}
 & & R/\mathfrak{p} & & \\
 & \nearrow \text{quotient} & & \searrow \text{localize} & \\
 R & & & & K(R/\mathfrak{p}) \cong R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \\
 & \searrow \text{localize} & & \nearrow \text{quotient} & \\
 & & R_{\mathfrak{p}} & &
 \end{array}$$

Proof:
exercise

3.3.5 Nakayama's Lemma

(Atiyah Michael in Comm alg has some interesting results that feel are worth putting in here: p.21 on fin. gen modules)

As mentioned at the beginning, localization let's us get geometric information on a local level. Nakayama's lemma will be one way we translate that information to a global level. In this section, we will restrict to finitely generated modules (in exercise ref:HERE, we will work with the infinite case).

Let's say M is a finitely generated module over R . If we were to localize M at a prime, we would get a local ring $R[D^{-1}]$. For simplicity, let's denote this ring G . As we've shown, G has only one maximal ideal m . Modding G by m , we get the field G/m . M over G/m is a vector-space, and so it has a basis. The heart of Nakayama's lemma is that this can be reversed: Using the basis of M over G/m , we can find a basis for M over G !

We will build this up in three steps, which are collectively called Nakayama's Lemma:

Definition 3.3.26: Jacobson Radical

Let R be a ring. Then the *jacobson radical* of R , denoted $\text{Jac}(R)$ is the intersection of all maximal ideals of R :

$$\text{Jac}(R) = \bigcap_{\substack{M \subseteq R \\ \text{maximal}}} M$$

(analogous to Frattini subgroup)

Proposition 3.3.27: Properties Of Jacobson Radical

Let $J \subseteq R$ be the Jacobson radical of a commutative ring R . Then:

1. If $I \subsetneq R$ is a proper ideal, then the ideal generated by I and J is also proper: $(I, J) \subsetneq R$
2. The jacobson radical is nilpotent (i.e. if $\text{rad}(J)^n = 0$ for some n)
3. $x \in J$ if and only if $1 - rx$ is a unit for every $r \in R$

Proof:

1. If I is maximal, then $J \subseteq I$ and we're done, and if not then $I \subseteq M$ for some maximal ideal so $(I, J) \subseteq M \subsetneq R$
2. Let J be the jacobson radical. By the D.C.C. the following chain stabilizes

$$J \supseteq J^2 \supseteq \dots \supseteq J^n = J^{n+1} = \dots$$

Consider the set of all annihilators of J^n :

$$I = \{x \in R \mid xJ^n = 0\}$$

I claim that $I = (1)$, which will show that $1J^n = J^n = 0$ (i.e. J is nilpotent). For the sake of contradiction, let's say it is not I . Then pick $I' \supsetneq I$ that is minimal (i.e. there is no ideal between I' and I). Such an ideal exists by the D.C.C. condition. Then by minimality, it must be that I' is generated by adding just one more element, that is

$$I' = I + Rx \quad x \in R, x \notin I$$

Thus, I'/I is a simple R -module by minimality (i.e. has no other ideals in it). However, it is easy to see that any simple R -module is of the form R/m for some maximal ideal m . Thus,

$$I'/I \cong R/m$$

multiplying both sides by m , we "kill" the right hand side and get

$$mI' \subseteq I$$

Thus $JI' \subseteq I$, so $xJ \in I$, and so by definition of I ,

$$xJ^{n+1} = 0$$

but by assumption $xJ^{n+1} = J^n$, but that implies $x \in I$. However, $x \notin I$, a contradiction.

3. Doing the contra-positive, let's say $x \notin \text{Jac}(R)$. Then $x \notin M'$ for some maximal ideal M' . By maximality, $M + (x) = (1)$, so $y + rx = 1$, implying $1 - rx = y \in M'$, showing that $1 - rx$ cannot be a unit (this does not show "forall $r \in R$ "). Conversely, if $1 - rx$ is not a unit, $1 - rx \in M$ for some M . However, $1, rx \notin M$, since if they were then $1 \in M$. Thus, $rx \notin \text{Jac}(R)$ for all r , but then $x \notin \text{Jac}(R)$.

Lemma 3.3.28: Nakayama's Lemma I

Let M be a finitely generated R -module, and suppose $\alpha \subseteq R$ is an ideal that is contained in all maximal ideals of R (i.e. $\alpha \subseteq \text{Jac}(R)$). Then:

$$\text{if } \alpha M = M \text{ then } M = \{0\}$$

Proof:

We will proceed by induction on the number of generators of M . If $n = 1$, then $\langle x_1 \rangle = M$. If $\alpha M = M$, we have that:

$$x_1 = a_1 x_1 \text{ for some } a_1 \in \alpha$$

we get that $(1 - a_1)x_1 = 0$, meaning $1 - a_1$ annihilates x_1 .

However, $(1 - a_1)$ is in fact a unit (as will be shown in a moment). Since it's a unit, there exists a b such that

$$0 = b \cdot 0 = b(1 - a_1)x_1 = x_1$$

Thus it must be that $x_1 = 0$, showing that $M = \{0\}$. So it's left to show that $1 - a_1$ is a unit. Let's say $(1 - a_1)$ was a proper ideal. Then it must lie in some maximal ideal. Since $a_1 \in \alpha$ and α is a subset of all maximal ideals, it must be that $1 - a_1 + a_1 = 1$ is in a maximal ideal, however that would make the ideal a non-proper ideal – a contradiction.

For the case of $n = k$, let $x_1, x_2, \dots, x_k$ generate M , and suppose $\alpha M = M$. Then

$$x_k = a_1 x_1 + a_2 x_2 + \dots + a_k x_k$$

so $(1 - a_k)x_k$ is generated by $k - 1$ elements. Thus by the induction hypothesis, those elements must in fact generate the zero-module, and so

$$(1 - a_k)x_k = 0$$

and by the same reasoning as before, it must be that $x_k = 0$, and so $M = \{0\}$, as we sought to show

If we restrict to $R = A$ being a local ring and $\alpha = m$ the maximal ideal of A , we will have the first step in constructing a basis for M over A :

Lemma 3.3.29: Nakayama's Lemma II

Let M be an A -module over a local ring A and let m be the maximal ideal, and let F be a submodule of M . Then:

$$\text{if } M = F + mM \text{ then } M = F$$

Proof:

Take $M/F = (F + mM)/F = mM/F$ so that $M/F = mM/F$. Then by lemma 3.3.28, $M/F = \{0\}$, so that means that M moded by F makes everything 0, and so

$$M = F$$

Finally, we put these two together to create a basis for a module over a local ring using a basis for a smaller module!

Lemma 3.3.30: Nakayama's Lemma III

Let M be a A -module over a local ring A and m the maximal ideal of A . If $x_1, x_2, \dots, x_n$ generated M/mM ($M \bmod mM$), then they also generators for M

Proof:

Let $F = \langle x_1, x_2, \dots, x_n \rangle = M/mM$. Then

$$M = F + mM$$

Then by lemma 3.3.29, $M = F$, and so M has $\langle x_1, x_2, \dots, x_n \rangle$ as generators, as we sought to show

Nakayama's lemma let's us reverse some constructions. As we saw in chapter ??, a [finitely generated?] projective module over a PID is free. Using Nakayama's lemma, we can show that a finitely generated module over a local ring is also free!

Theorem 3.3.31: Projective Module over Local Ring

Let M be a finitely generated projective module over a local ring A . Then M is free. In particular, If $\{x_1, \dots, x_n\}$ are elements of M who's residue class $\{\bar{x}_1, \dots, \bar{x}_1\}$ form a basis for M/mM over A/m , then $x_1, \dots, x_n$ is a *basis* for M over A .

Furthermore, if $\bar{x}_1, \dots, \bar{x}_n$ are linearly independent over A/m , then this collection can be completed to a basis for M over A

Proof:

First, Note that $M/\mathfrak{m}M$ over $A/\mathfrak{m}$ is indeed a vector space with action $(a + \mathfrak{m})(m + \mathfrak{m}M) = (am + \mathfrak{m}M)$ being well-defined.

Since M is finitely generated, let $\{x_1, x_2, \dots, x_n\}$ be the generating set. Let F be a free module with basis $e_1, e_2, \dots, e_n$, and consider the homomorphism $f : F \rightarrow M$, $f(e_i) = x_i$. We'll show that f is in fact an isomorphism. Since $x_1, \dots, x_n$ generate $M \bmod \mathfrak{m}M$, by Nakayama's lemma III, they generate M and so f is surjective. Next, since M is projective, f splits, i.e., $F = P_0 \oplus P_1$ where $P_0 = \ker(f)$ and P_1 and $P_1 \cong M$ through f . We'll show $P_0 = 0$.

Take some $\sum a_i x_i \in \ker(f)$ so that $\sum a_i x_i = 0$. Then reducing to the mod, we have $\sum a_i x_i + \mathfrak{m}M = 0 + \mathfrak{m}M = \sum (a_i \mathfrak{m}) \cdot (x_i + \mathfrak{m}M) = 0 + \mathfrak{m}M$. Since the $\bar{x}_i$ form a basis for $M/\mathfrak{m}M$ over

$A/\mathfrak{m}A$, it must be that $a_i \in \mathfrak{m}$. Thus, we have $\sum a_i e_i \in \mathfrak{m}F$. Therefore:

$$\ker(f) \subseteq \mathfrak{m}F = \mathfrak{m}\ker(f) \oplus \mathfrak{m}M$$

Thus, $\mathfrak{m}\ker(f) = \ker(f)$. Since $\ker(f)$ is the direct summand of finitely generated modules, it is itself finitely generated. Thus, by Nakayama's lemma, $\ker(f) = 0$, and so f is an isomorphism! Therefore, we have $F \cong M$, and so M is free!

Notice that this immediately implies that the $x_1, \dots, x_n$ are lineally independent since they form a basis for M over A (since f is an isomorphism and $f(e_i) = x_i$). Finally, if we extend $\bar{x}_1, \dots, \bar{x}_r$ to $\bar{x}_1, \dots, \bar{x}_r, \dots, \bar{x}_n$ to be a basis for $M/\mathfrak{m}M$, then we just showed they will indeed form a basis for M .

A cool result of this theorem is a way to show some properties f based on f on the respective local modules. Let M be module over a local ring A . Then there is an associated module $M(\mathfrak{m}) = M/\mathfrak{m}M$. Then $\text{---}(\mathfrak{m})$ induces a functor mapping M to $M(\mathfrak{m})$ and

$$f : M \rightarrow N \quad \mapsto \quad f_{(\mathfrak{m})} : M(\mathfrak{m}) \rightarrow N(\mathfrak{m})$$

Notice that if f is surjective, then $f_{(\mathfrak{m})}$ is surjective. The magic of Nakayama's lemma is that the converse is true too!

Proposition 3.3.32: Local Homomorphisms

Let M, N be finitely generated modules over a local ring A and $f : M \rightarrow N$ be an A -module homomorphism. Then:

1. If f is surjective if and only if $f_{(\mathfrak{m})}$ is surjective
2. Assume f is injective. Then if $f_{(\mathfrak{m})}$ is surjective, f is an isomorphism
3. If M and N are free, f is injective (resp. an isomorphism) if and only if $f_{(\mathfrak{m})}$ is injective (resp. an isomorphism)

Found this page that proves the results quite nicely: <http://homepage.divms.uiowa.edu/~acwood/files/notes/nakayamaslemma.pdf>

Proof:

1. Let $f : M \rightarrow N$ be an R -module homomorphism and let the induced $f_{(\mathfrak{m})}$ be surjective so that $f(M \text{ mod } \mathfrak{m}M) = N \text{ mod } \mathfrak{m}N$. Since N is finitely generated, so is any quotient of it, and so $N \text{ mod } \mathfrak{m}N$ is finitely generated by $\{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n\}$. By Nakayama's lemma, $\{x_1, x_2, \dots, x_n\}$ generated N . Since $x_1, \dots, x_n \in f(M)$, f is also surjective. More consisely, since $f_{(\mathfrak{m})}$ is surjective, $N = \text{im}(f) + \mathfrak{m}N$. Thus, by Nakayama's lemma, $N = \text{im}(f)$.
2. here
3. Assume M and N are free and $f_{(\mathfrak{m})}M/\mathfrak{m}M \rightarrow N/\mathfrak{m}N$ is injective. Since M is free, let $\{x_1, \dots, x_n\} \subseteq M$ so that $\{\bar{x}_1, \dots, \bar{x}_n\}$ is a basis for $M/\mathfrak{m}M$. Then by theorem 3.3.31, $x_1, \dots, x_n$ forms a basis for M . Suppose now that $f(a_1x_1 + \dots + a_nx_n) = 0$. By definition of $f_{(\mathfrak{m})}$, $f_{(\mathfrak{m})}(\bar{a}_1\bar{x}_1 + \dots + \bar{a}_n\bar{x}_n) = \bar{0}$. Since $f_{(\mathfrak{m})}$ is injective, we have $\bar{a}_1\bar{x}_1 + \dots + \bar{a}_n\bar{x}_n = \bar{0}$, and since $\bar{x}_1, \dots, \bar{x}_n$ is a basis, we have that the coefficients are all $\bar{0}$, that is $a_i \in \mathfrak{m}$.

To show that $\ker(f) = 0$, we'll use Nakayama's lemma, so it suffices to show that $\ker(f) \subseteq \mathfrak{m}\ker(f)$. By what we've just shown, we have $\ker(f) \subseteq \mathfrak{m}M$. If $f_{(\mathfrak{m})}$ were surjective, then this proof is rather quick. By part (1), we have that f is surjective. Thus the following short exact sequence splits:

$$0 \rightarrow \ker(f) \rightarrow M \rightarrow N \rightarrow 0$$

since N is free, So $M \cong \ker(f) \oplus N$, and $\ker(f) \subseteq \mathfrak{m}M \cong \ker(f) \oplus N$, and so $\ker(f) \subseteq \mathfrak{m}\ker(f)$, and so by Nakayama's lemma $\ker(f) = 0$.

If $f_{(\mathfrak{m})}$ is not surjective, then we need to do a little more work. We'll take advantage of the basis lifting property we proved in the previous theorem. Let $\{\bar{x}_1, \dots, \bar{x}_n\}$ be a basis for $M/\mathfrak{m}M$ over $A/\mathfrak{m}A$ which induces a basis $\{x_1, \dots, x_n\}$ for M over A . $\langle f_{(\mathfrak{m})}(\bar{x}_1), \dots, f_{(\mathfrak{m})}(\bar{x}_n) \rangle = \text{im}(f_{(\mathfrak{m})})$ is a submodule of $N/\mathfrak{m}N$ which have that generating set as a basis (since $f_{(\mathfrak{m})}$ is injective). Thus, if:

$$\bar{a}_1 f_{(\mathfrak{m})}(\bar{x}_1) + \dots + \bar{a}_n f_{(\mathfrak{m})}(\bar{x}_n) = 0$$

then since f is a homomorphism, we have

$$f_{(\mathfrak{m})}(\bar{a}_1 \bar{x}_1 + \dots + \bar{a}_n \bar{x}_n) = 0$$

and since these elements form a basis and f is injective, $\bar{a}_i = 0$ for all i . Now, extend this to a basis for $N/\mathfrak{m}N$ over $A/\mathfrak{m}A$:

$$\{f_{(\mathfrak{m})}(\bar{x}_1), \dots, f_{(\mathfrak{m})}(\bar{x}_n), \bar{y}_1, \dots, \bar{y}_m\}$$

Since N is free, this basis lifts to a basis for N over A , in particular:

$$\{f(x_1), \dots, f(x_n), y_1, \dots, y_m\}$$

since $f_{(\mathfrak{m})}(\bar{x}_i) = f_{(\mathfrak{m})}(x_i + \mathfrak{m}M) = f(x_i) + \mathfrak{m}N$. With this, we can now get our conclusion: since the $f(x_i)$ are linearly independent, we have that:

$$a_1 f(x_1) + \dots + a_n f(x_n) = f(a_1 x_1 + \dots + a_n x_n) = 0$$

then it must be that the $a_i = 0$ for all the i , but then this is the condition for a function to be injective, and so f is injective, as we sought to show

commented region here:

Proposition 3.3.33

Let P and P' be A -module over a local ring A . Then $P \cong P'$ if and only if $P/\mathfrak{m}P \cong P'/\mathfrak{m}P'$.

Proof:

Any finitely generated projective module over a local ring is free, and so apply the previous proposition

There is a re-formulation of Nakayama's lemma using the tensor product, however I omitted it for

now.

(Another application of Nakayama's Lemma)

Recall that if R is a finite integral domain, then it is necessarily a field (??). Naturally, this is not necessarily the case for infinite rings ($\mathbb{Z}$ being the prototypical example). We may thus take $Q = \text{Frac}(R)$, and consider the R -module Q

Corollary 3.3.34: Infinite Generation Of Ring Of Fractions

Let $Q = \text{Frac}(R)$ be the ring of fractions of R for an infinite ring R . Then considered as an R -module, Q is infinitely generated.

Proof:

Let's assume that Q was a finitely generated R module. Then $Q_{\mathfrak{m}}$ is a finitely generated $Q_{\mathfrak{m}}$. But then $\mathfrak{m}_{\mathfrak{m}}Q = Q$, and so Nakayama's lemma implies $F = 0$, a contradiction.

Recall that when dealing with infinite functions, we may have some strange results, like a surjection $f : M \rightarrow M$ that is not an injection. Nakayama's lemma insures we get some sanity here:

Corollary 3.3.35: Surjective Implies Injective

Let M be a finitely generated R -module. Then if $f : M \rightarrow M$ is a surjective R -module homomorphism, it is injective.

Note how no assumption about R or M being Noetherian!!

Proof:

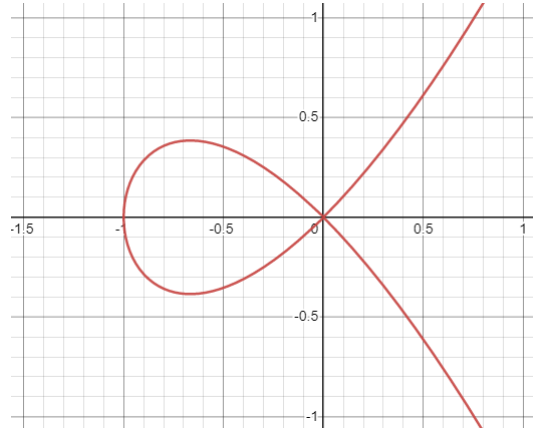
Need a slightly more general version of Nakayama's lemma, see comment in document, really cool!

Completion

The next type of construction we'll do is called a *completion*. These will be in the appropriate sense “stronger” than localizations, which will result in them being more restrictive, though also more powerful. A completion of a ring in many ways parallels the completion of the rational numbers $\mathbb{Q}$ to the real numbers $\mathbb{R}$. In fact, defining the appropriate metric, this process is a special case of the Cauchy-sequence completion of the rational numbers to the reals. Namely, the completion of a ring is the equivalence class of all Cauchy-sequences which converge to the same point. Due to the topological background needed for this perspective, we shall relegate the equivalence of what shall be explicated in this section and Cauchy-sequences to section ??.

There were rings that we have observed earlier that are completions of simpler rings; we will use completions to construct rings like the ring of *formal power series* $R[[x]]$ from $R[x]$, the ring $R[x_1, x_2, \dots]$, the ring with *infinitely many monomial* from $R[x_1, \dots, x_n]$, and the *p-adic integers* $\mathbb{Z}_p$ from $\mathbb{Z}$.

Completions will be handy for us in many ways. One important use that we will see when looking into algebraic geometry is looking at the “singularities” of schemes (or more particularly algebraic curves). Singularities will be points that don't behave as nicely as most other points, for example in the following image (which is the zero set of the equation $y^2 - x^2(x + 1) = 0$) does not behave well at $(0, 0)$:

Figure 4.1: zero set of $y^2 - x^2(x+1) = 0$

This may seem similar to the geometric notion of localization, and this is indeed the case. Indeed, in the case of $\mathbb{Z}_{(p)}$ the p -adic integers are the completion of this ring, namely $\mathbb{Z}_p$ is the completion of $\mathbb{Z}_{(p)}$.

Completions shall also play a very important role in algebraic number theory. In this book, a more “classical” approach to algebraic number theory was taken, which shall give the reader great insights on the fundamental of algebraic number theory. However, a large part of *class number theory* shall rely on special completions of rings.

In this section, topological notion’s will be introduced, and so an acquaintance of the material in appendix ?? is recommended. It should be noted that the topologies that will be discussed will be very different from that of Euclidean topologies, a new notion of size that is rather different from Euclidean distance shall be developed. The reader that is familiar with the construction of the real numbers using Cauchy sequences shall find it much easier to follow this section.

4.1 Constructing Completions

We will first define what is a “sequence”. Let’s say we have a collection of countable rings $R_1, R_2, \dots, R_n$ and homomorphisms:

$$\dots \xrightarrow{f_3} R_3 \xrightarrow{f_2} R_2 \xrightarrow{f_1} R_1$$

Then a sequence, or a *coherent sequence* is an element $(x_1, x_2, \dots) \in \prod_{i=1}^{\infty} R_i$ where:

$$f_i(x_{i+1}) = x_i$$

If we take the special case of $I \subseteq R$ being an ideal and

$$\dots \xrightarrow{f_3} R/(I^3) \xrightarrow{f_2} R/(I^2) \xrightarrow{f_1} R/I$$

then we will define a form of completion from these maps. For ease of notation, the sequence is usually written in the reverse order, as the reader shall understand once examples will be presented

Definition 4.1.1: Completion Of A Ring

Let R be a ring and $I \subseteq R$ be an ideal, and:

$$R/I \xleftarrow{f_1} R/(I^2) \xleftarrow{f_2} R/(I^3) \leftarrow \dots$$

Then:

$$\hat{R} := \left\{ (a_1, a_2, \dots) \in \prod_{i=1}^{\infty} R_i \mid f_i(a_{i+1}) = a_i \right\}$$

which is often denoted as:

$$\varprojlim R/I^n := \hat{R}$$

and is called the *limit* (or in older textbook the *inverse limit* or *projective limit*). The collection $\{R/I^n\}$ is called the *inverse system*, with the function usually taken as implicit

A couple of remarks are in order:

1. There is a natural topology and a metric in which these elements can actually be thought of as Cauchy sequences and $\hat{R}$ is indeed the completion of R . If $\{R/\mathfrak{a}^n R\}$ is an inverse system with ideal $\mathfrak{a} \subseteq R$, and any $x \in R$, define a neighborhood basis (or local basis) $\{x + \mathfrak{a}^n R\}_{n=1}^{\infty}$. Then this forms a basis for a topology. This topology is called the *$\mathfrak{a}$ -adic topology*, U is open in this topology if and only if for each $x \in U$, there exists a positive integer n such that

$$x + \mathfrak{a}^n R \subseteq U$$

Then R is Hausdorff if and only if $\bigcap \mathfrak{a}^n R = 0$. If this is the case, define the metric $d(x, y) = 2^{-n}$ whenever $x \neq y$ where n is an integer such that $x - y \in \mathfrak{a}^n R - \mathfrak{a}^{n-1} R$. The constant 2 does not matter, it can be replaced with any $C > 1$ and it will give the same topology.

This exact same process can be repeated on an arbitrary R -module, where the inverse system would be $\{M/\mathfrak{a}^n M\}$, and for a given ideal $I \subseteq R$, we would define the I -adic topology on M .

2. This is a special case of a more general construction of a *limit* in the category of **Ring**. Namely, it is the limit of a special form of the diagram:

$$\dots \rightarrow \bullet \rightarrow \bullet \rightarrow \bullet$$

where each arrow has the coherence condition stated above.

3. By how we've defined it, any inverse system will be surjective. Such systems are sometimes *surjective systems*.
4. If R is Noetherian and integral, then $\bigcap_n I^n = 0$, and hence $R \subseteq \hat{R}$. For those who shall read appendix ??, this shows that integral Noetherian domains are Hausdorff in the I -adic topology.

Example 4.1.2: Completion of Rings

1. Let $R = k[x]$ and $I = (x)$. Then:

$$\varprojlim (k[x]/(x)^n) = k[[x]]$$

To see this, consider:

$$\cdots k[x]/(x^3) \rightarrow k[x]/(x^2) \rightarrow k[x]/(x) = k$$

Notice that in each $k[x]/(x^k)$, can be thought of the collection of all polynomial of degree up to $k - 1$. Hence, a coherent sequence $(a_1, a_2, \dots)$ can be thought of as an “infinitely long polynomial” and would usually be denoted

$$\sum_k a_k x^k$$

Similarly, the reader may verify that $\widehat{k[x, y]} = k[[x, y]]$ with $I = (x, y)$

2. Let $R = \mathbb{Z}$ and $I = (10)$. Then $\varprojlim \mathbb{Z}/(10)^n$ is represented by all infinitely long numbers. Just like before, each $\mathbb{Z}/(10)^k$ can be thought of as holding information up to the $k - 1$ th position. So if we take:

$$\mathbb{Z}/(10) \leftarrow \mathbb{Z}/(100) \leftarrow \mathbb{Z}/(1000) \leftarrow$$

and construct a coherent chain, we may chose elements like 5, 25, 425, ... and so on. The reader may see that this would represent an infinitely long number. The resulting ring is often denoted $\mathbb{Z}_{10}$ and is called the 10-adic integers. Any element of $\mathbb{Z}_{10}$ could formally be represented as

$$\sum_k a_k 10^k$$

3. Let $R_n = k[x_1, \dots, x_n]$ be the polynomial in n variables. Then the completion is $k[x_1, \dots, x_n, \dots]$

Completions are a way to construct rings. It would be nice if exactness interact nicely with it. If $\{A_n\}$, $\{B_n\}$, $\{C_n\}$ are three inverse systems where:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A_{n+1} & \xrightarrow{\psi} & B_{n+1} & \xrightarrow{\varphi} & C_{n+1} \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & A'_n & \xrightarrow{\psi'} & B'_n & \xrightarrow{\varphi'} & C'_n \longrightarrow 0 \end{array}$$

Then we'll denote it as $0 \rightarrow \{A_n\} \rightarrow \{B_n\} \rightarrow \{C_n\} \rightarrow 0$. Then we naturally get the induced homomorphisms:

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n \rightarrow 0$$

however this is not always exact. The following proposition shows us when it is:

Proposition 4.1.3: Limits and Exactness for inverse Systems

Let $0 \rightarrow \{A_n\} \rightarrow \{B_n\} \rightarrow \{C_n\} \rightarrow 0$ be an exact sequence of inverse systems. Then:

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n$$

is always exact. If $\{A_n\}$ is a surjective system, then:

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n \rightarrow 0$$

is always exact

Hence, when working with completions induced by surjective systems, the completion functor $\widehat{(-)}$ preserves injective and surjective functions¹.

Proof:

This is an application of Snakes lemma. Let $A = \prod_{n=1}^{\infty} A_n$, and

$$d^A : A \rightarrow A \quad d^A(a_n) = a_n - f_{n+1}(a_{n+1})$$

Then by construction

$$\ker d^A = \varprojlim A_n$$

We may define B, C via d^B and d^C respectively. Now, as we have an exact sequence of inverse systems, we get the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow d^A & & \downarrow d^B & & \downarrow d^C \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \end{array}$$

Then by ?? we get:

$$0 \rightarrow \ker d^A \rightarrow \ker d^B \rightarrow \ker d^C \rightarrow \operatorname{coker} d^A \rightarrow \operatorname{coker} d^B \rightarrow \operatorname{coker} d^C \rightarrow 0$$

Finally, if the system is a surjective system, we can solve inductively the equation:

$$x_n - f_{n+1}(x_{n+1}) = a_n$$

to show that d^A is surjective, completing the proof.

The following result shall be stated as it gives a very fascinating result, however for the full background the reader should read section ??:

¹In part ??, this shall imply that the completion functor has trivial cohomology, while general inverse systems shall have non-trivial cohomology

Corollary 4.1.4: Completion of Group Exact Sequences

Let $0 \rightarrow G' \rightarrow G \xrightarrow{p} G'' \rightarrow 0$ be an exact sequence of groups. Let G have a topology given by the subgroups $\{G_n\}$, and induce the topology on G', G'' via the subgroups $\{G'_n \cap G_n\}$ and $\{pG_n\}$. Then:

$$0 \rightarrow \hat{G}' \rightarrow \hat{G} \rightarrow \hat{G}'' \rightarrow 0$$

is exact

Proof:

Apply proposition 4.1.3 to

$$0 \rightarrow \frac{G'}{G' \cap G_n} \rightarrow \frac{G}{G_n} \rightarrow \frac{G''}{pG_n} \rightarrow 0$$

Corollary 4.1.5: Quotient in Completion

Let $\hat{G}_n \leq \hat{G}$. Then:

$$\hat{G}/\hat{G}_n \cong G/G_n$$

Proof:

exercise

As a consequence, in particular interest is the case of $\mathbb{Z}_{10}$ where

$$\mathbb{Z}_{10} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_5$$

Hence $\mathbb{Z}_{10}$ is decomposable to smaller rings. These rings are also in many ways nicer: notice that as a consequence of the above isomorphism that $\mathbb{Z}_{10}$ has zero divisors (any product of two non-trivial rings has zero-divisors)², showing that in general the completion may introduce zero-divisors, while $\mathbb{Z}_2$ and $\mathbb{Z}_5$ do not introduce zero-divisors. Furthermore, $\mathbb{Z}_p$ introduces more units (this was more evident with $k[[x]]$ where the units were explicitly found in ??). For this reason, we would concentrate on the p -adic numbers $\mathbb{Z}_p$. Any element of $\mathbb{Z}_p$ could be represented as

$$\sum_k a_k p^k$$

We may generalize the result for finding units in completions:

Lemma 4.1.6: Units In Completion

Let R be a ring and $\hat{R}$ a completion of R via p . Then if $u \in R/p^n$ is a unit, $u \in R/p^{2n}$ is a unit.

²Can you find two explicit elements?

Proof:

Let u be a unit so $1 + us \in p^n$. Squaring, we get:

$$1 + u(2s + us^2) \in p^{2n}$$

Hence, it is a unit in R/p^{2n} .

The reader should use this to show that u is a unit in $\hat{R}$. This also generalizes ??, for this directly shows that any power series with nonzero constant is a unit. In the case where $p = \mathfrak{m}$ is a maximal ideal (as in the case for p -adic integers and $k[[x]]$), we get the following result:

Lemma 4.1.7: Completion At Maximal Ideals

Let R be a ring and $\mathfrak{m}$ a maximal ideal. Then $\varprojlim R/\mathfrak{m}^n$ is a local ring, that is it contains a single maximal field.

Proof:

Notice that all elements in R that are not in $\mathfrak{m}$ are units in $R/\mathfrak{m}$. Thus, by lemma 4.1.6 they are units in $\varprojlim R/\mathfrak{m}^n$.

Thus, there is a natural map:

$$R_{\mathfrak{m}} \rightarrow \varprojlim R/\mathfrak{m}^n$$

where $R_{\mathfrak{m}}$ is the localization. Recall that in $R_{\mathfrak{m}}$, the elements are constructed by taking formal inverses, hence completion at a maximal ideal shows a “dual” way of creating units by finding them in some way “at infinity”. Note that it is important that $\mathfrak{m}$ be maximal, for this will fail if an ideal is only prime (however, one may localize the ring at the prime ideal to make it maximal and then take the completion).

When working with localization, we saw that a localization of an integral domain is an integral domain. This is no longer true for completion:

Example 4.1.8: Completion Looses Integral Domain

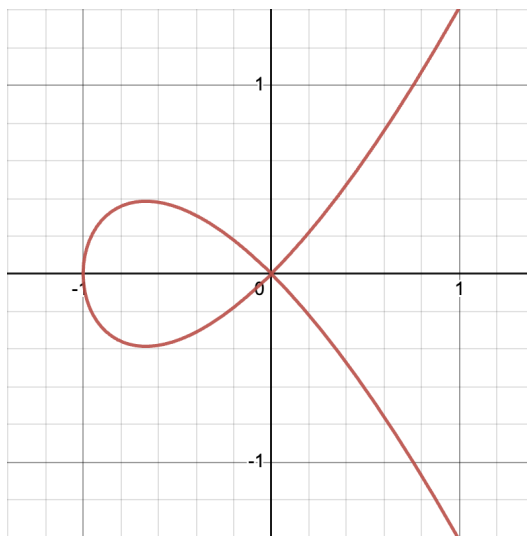
Let

$$R = \frac{k[x, y]}{(y^2 - x^3 - x^2)}$$

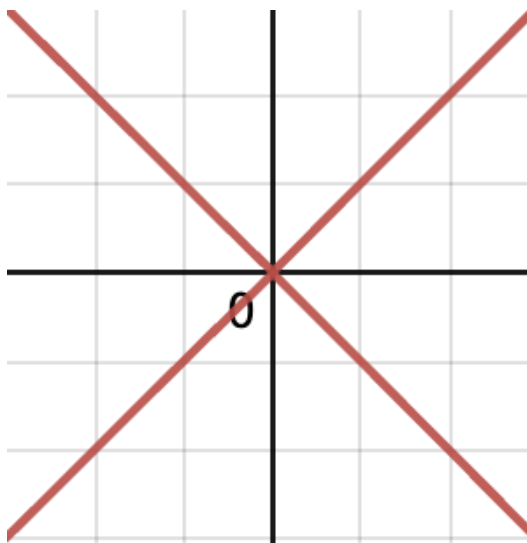
with k a field of characteristic 0. Then completing this ring, since completion is exact, we get:

$$\hat{R} = \frac{k[[x, y]]}{(y^2 - x^3 - x^2)} \cong \frac{k[[x, y]]}{(y - x\sqrt{1+x})(y + x\sqrt{1+x})}$$

where $\sqrt{1+x}$ is the short-hand for the power-series associated to $\sqrt{1+x}$. Then it is easy to see that this ring has zero-divisors. This shall eventually be made rather clear when a geometric perspective is taking. The ring R shall in a concrete sense have the curve $y^2 = x^3 + x^2$ associated to it:

Figure 4.2: $y^2 = x^3 + x^2$

This curve shall be shown to be *irreducible* in a concrete way. The localization shall represent this curve minus some finite amount of points. The completion at (x, y) shall represent “zooming in” at the point (x, y) , at which point geometric object shall roughly correspond to:

Figure 4.3: Approximate Geometric picture corresponding to $\varprojlim R/(x, y)^n$

Then this shall be shown to be decomposable in a rigorous sense into two components (the two strokes of the “x” in the above picture).

Definition 4.1.9: Complete Ring and Module

Let R be a ring and M an R -module. Then R is said to be *complete* if

$$R \cong \hat{R}$$

And hence $R \cong \varprojlim R_n$. M is said to be *complete* if

$$M \cong \hat{M}$$

where $\hat{M}$ is given by taking the limit of $G = M$ and $G_n = R_n \cdot M$. In this case, the ring is complete and we can say it is an $\hat{R}$ -module.

Show that if we take $R_n = \mathfrak{a}^n$ for an ideal, then $f(\mathfrak{a}^n M) = \mathfrak{a}^n f(M) \subseteq \mathfrak{a}^n N$, and so any R -module homomorphism lifts to an $\hat{R}$ -module homomorphism, and similarly in the more general case.

4.2 Filtrations

The following construction is a more general version of an inverse system that drops many of the algebraic properties and is a more “pure topological” definition. It is being presented here for its use in the study of graded algebras, as well as its use when defining *spectral sequences* in chapter ??

Definition 4.2.1: Filtration

Let M be an R -module. Then a *filtration* of M is an infinite chain:

$$M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \supseteq \cdots$$

where each M_n is a submodule of M . The chain is usually denoted (M_n) . If furthermore $\mathfrak{a} \subseteq R$ is an ideal, then an $\mathfrak{a}$ -filtration of M is a filtration st

$$\mathfrak{a}M_n \subseteq M_{n+1} \quad \forall n \in \mathbb{N}$$

A filtration is called *stable* if for some $N \in \mathbb{N}$, for all $N \geq n$, $M_N = M_{N+1}$. A filtration is called $\mathfrak{a}$ -stable if for some $N \geq n$, $\mathfrak{a}M_N = M_{n+1}$.

An $\mathfrak{a}$ -stable filtration can be thought of as the generalization of a graded module which is in some way finite. The following lemma will be important to compare two $\mathfrak{a}$ -stable filtrations:

Lemma 4.2.2: Comparing Stable $\mathfrak{a}$ -Filtrations

Let $(M_n), (M'_n)$ be stable $\mathfrak{a}$ -filtrations of M . Then they have bounded difference, that is, there exists an integer n_0 such that $M_{n+n_0} \subseteq M'_n$ and $M'_{n+n_0} \subseteq M_n$ for all $n \geq 0$.

This will imply that all stable $\mathfrak{a}$ -filtrations determine the same topology on M (The $\mathfrak{a}$ -topology)

Proof:

Let $M'_n = \mathfrak{a}M$. Then as $\mathfrak{a}M_N \subseteq M_{n+1}$, $\mathfrak{a}^n M \subseteq M_n$. Furthermore, $\mathfrak{a}M_n = M_{n+1}$ for all $n \neq n_0$ for appropriate n_0 , and so $\mathfrak{a}^n M_{n_0} \subseteq \mathfrak{a}M$, completing the proof.

The key use of this lemma comes when defining the α -adic topology given in appendix ??, namely all stable $\mathfrak{a}$ -filtrations produce the same topology.

Exercise 4.2.1

1. Show that $\mathbb{Z}_p \cap \mathbb{Q} \cong \mathbb{Z}_{(p)}$
2. Show that $\hat{\hat{R}} \cong \hat{R}$

4.3 More on Graded Algebras

Graded algebras were initially explored in section ??. In this section, we expand further on graded-algebras, explore their relation to the Noetherian property, and look at some important associated graded algebras given an ideal I or module M . This exploration shall bear fruitful results that will be important in the development of *projective algebraic geometry* and *homological algebra*.

First, recall that a ring R is a graded ring if $R = \bigoplus_i R_i$ where $R_i R_j \subseteq R_{i+j}$, which makes R into an R_0 -algebra, and each R_i is an R_0 -module. A graded R -module is a module $M = \bigoplus_0^\infty M_i$ such that $R_m M_n \subseteq M_{m+n}$. If M is also a graded ring, then M is an R -graded algebra. Elements of M_n are called *homogeneous*, and an ideal $I \subseteq R$ is called homogeneous if $I = \bigoplus_i^\infty (R_i \cap I)$. A ring or R -module homeomorphism is called graded if it respects the graded structure ($f(M_n) \subseteq N_n$). Each graded ring has an ideal $R_+ = \bigoplus_{n>0} R_n$. The elements of R_n for each n are called the *homogeneous elements*.

Proposition 4.3.1: Noetherian Graded Rings

Let R be a graded ring. Then the following are equivalent:

- ($\Rightarrow$) R is Noetherian
- ($\Rightarrow$) R_0 is Noetherian and R is a finitely generated R_0 algebra

Proof:

- (1) $\Rightarrow$ (2) As $R_0 \cong R/R_+$, R_0 is Noetherian. As $R_+ \subseteq R$ is an ideal of R , it is finitely generated, say $x_1, \dots, x_s$. Then these elements can be used to construct homogeneous s element, let's use the same label and say they are of order $k_1, \dots, k_s$ (all necessarily order greater than 0). Let $R' \subseteq R$ be the ring generated as an R_0 algebra over the homogeneous elements $x_1, \dots, x_s$. Then I claim that $R = R'$, which will be the proof. First, let's show $A_n \subseteq A'$ using induction. It is clear if $n = 0$. For $n > 0$, taking $y \in A_n$, as $y \in A_+$, y is a linear combination of the x_i

$$y = \sum_i^s a_i x_i$$

with $a_i \in A_{n-k_i}$ (and $A_m = 0$ if $m < 0$). Then by the inductive hypothesis, each a_i is a polynomial in the x 's with coefficients in A_0 . Hence, by the principle of induction, the same holds for y , we have that $A_n \subseteq A'$. As $A' \subseteq A$ by definition, we have $A = A'$, as we sought to show

(2) $\Rightarrow$ (1) Follow's from HBT

Any ring R induces a graded ring by taking $\mathfrak{a} \subseteq R$ and taking $R^* = \bigoplus_n^\infty \mathfrak{a}^n$. Similarly, any R -module M induces a graded R^* -module $M^* = \bigoplus_n M_n$ where M_n is the $\mathfrak{a}$ -filtration. The “canonical” grading comes from $S = R[x_1, \dots, x_n]$ with $\mathfrak{a} = (x_1, \dots, x_n)$ which will produce the homogeneous grading of S . In the case where R is Noetherian and $\mathfrak{a}$ is finitely generated (say by $x_1, \dots, x_s$), then $R^* = R[x_1, \dots, x_s]$ and is Noetherian by proposition 4.3.1.

Proposition 4.3.2: Topology of Noetherian Graded Module

Let R be a Noetherian ring, M a finitely generated R -module, (M_n) an $\mathfrak{a}$ -filtration of M . Then the following are equivalent:

- ($\Rightarrow$) M^* is a finitely generated R^* -module
- ($\Rightarrow$) The filtration (M_n) is stable

Proof:

This is about creating the right conditions to see the filter is stable. Take $Q_n = \bigoplus_{r=0}^n M_r$. As each M_r is finitely generated, Q_n is finitely generated. Then Q_n forms a subgroup of M^* but not in general a R^* -submodule. We may however use it to generate one:

$$M_n^* = M_0 \oplus \dots \oplus M_n \oplus \mathfrak{a}M_n \oplus \mathfrak{a}^2M_n \oplus \dots \oplus \mathfrak{a}^rM_n \oplus \dots$$

Now, as Q_n is finitely generated as an R -module, M_n^* is finitely generated as an R^* -module, and the M_n^* form an ascending chain whose unions is M^* .

Finally, we now have that as R^* is Noetherian, M^* is finitely generated as an R^* -module if and only if the chain stops ($M^* = M_{n_0}^*$) if and only if the filtration is table.

Corollary 4.3.3: Graded Noetherian then Component Noetherian

Let R be graded and Noetherian, and suppose M is a finitely generated graded R -module. Then M_n is a finitely-generated R_0 -module for all $n \in \mathbb{N}$

Proof:

Let

$$d_j = \deg m_j$$

Then every element of $m \in M_n$ can be written as:

$$m = \sum_j f_j(x)m_j \quad f_j(x) \in R$$

where f_j is homogeneous of degree $n - d_j$ (and is zero if $n < d_j$). Hence, M_n is finitely generated

as an R_0 -module generated by all $g_j(x)m_j$ where $g_j(x)$ is a monomial in x_i of total degree $n - d_j$, as we sought to show.

4.4 Artin-Rees Lemma

This is an excellent video: [here](#)

The Artin-Rees lemma is a central result that will allow us to compare the induced and $\mathfrak{a}$ -adic topology on a submodule, giving us insight on their topologies.

Theorem 4.4.1: Artin-Rees Lemma

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R -module, and (M_n) a stable $\mathfrak{a}$ -filtration of M . Then if $M' \subseteq M$ is a submodule of M , $(M' \cap M_n)$ is a stable $\mathfrak{a}$ -filtration of M'

Proof:

As $\mathfrak{a}(M' \cap M_n) \subseteq \mathfrak{a}M' \cap \mathfrak{a}M_n \subseteq M' \cap M_{n+1}$, $(M' \cap M_n)$ is an $\mathfrak{a}$ -filtration, and so it defines a graded R^* -module which is a submodule of M^* , and so it is finitely generated, letting us apply by proposition 4.3.2 to completing the proof.

Corollary 4.4.2: Graded Sub-module, Computational Shortcut

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R -module, and (M_n) a stable $\mathfrak{a}$ -filtration of M . Then there exists a k such that

$$(\mathfrak{a}^n M) \cap M' = \mathfrak{a}^{n-k} ((\mathfrak{a}^k M) \cap M')$$

for all $n \geq k$

Proof:

exercise.

Theorem 4.4.3: Submodule Adic Topology Equivalence

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R -module, $M' \subseteq M$ a submodule. Then the filtrations

$$\mathfrak{a}^n M' \quad \text{and} \quad (\mathfrak{a}^n M) \cap M'$$

have bounded difference, hence the particular $\mathfrak{a}$ -topology of m'' coincides with the induced topology given by the $\mathfrak{a}$ -topology of M

Proof:

Direct result of corollary 4.4.2 and theorem 4.4.1.

Corollary 4.4.4: Module Completion Preserves Exactness

Let

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

be a short exact sequence of finitely generated R -modules over a Noetherian ring R . Then if $\mathfrak{a} \subseteq R$ is an ideal, the $\mathfrak{a}$ -adic completion functor preserves exactness:

$$0 \rightarrow \hat{M}' \rightarrow \hat{M} \rightarrow \hat{M}'' \rightarrow 0$$

Proof:

Theorem 4.4.3 provides the condition needed to use corollary 4.1.4.

Next, as we have a natural homomorphism $R \rightarrow \hat{R}$, we can think of $\hat{R}$ as an R -algebra, and hence we have an extension of scalars of an R -module M given to an $\hat{R}$ -module $\hat{R} \otimes_R M$. This gives a natural sequence of maps that we may want to understand:

$$\hat{R} \otimes_R M \rightarrow \hat{R} \otimes_A \hat{M} \rightarrow \hat{R} \otimes_{\hat{R}} \hat{M} = \hat{M}$$

In general, these maps are neither injective nor surjective. Under nicer conditions, we have a better control:

Proposition 4.4.5: Module Completion as Tensor Product

Let R be any ring. Then if M is finitely generated,

$$\hat{R} \otimes_R M \twoheadrightarrow \hat{M}$$

is surjective. If furthermore R is Noetherian, then:

$$\hat{R} \otimes_R M \cong \hat{M}$$

Proof:

Let $F \cong R^n$ so that $M \cong F/N$ for appropriate N . The conclusion can be drawn from the following commutative diagram:

$$\begin{array}{ccccccc} \hat{R} \otimes_R N & \longrightarrow & \hat{R} \otimes_R F & \longrightarrow & \hat{R} \otimes_R M & \longrightarrow & 0 \\ \downarrow \gamma & & \downarrow \beta & & \downarrow \alpha & & \\ 0 & \longrightarrow & \hat{N} & \longrightarrow & \hat{F} & \xrightarrow{\delta} & \hat{M} \longrightarrow 0 \end{array}$$

where the top line is given by the right-exactness of the tensor product, and the bottom line is given by corollary 4.1.4. Note that as completion commutes with finite direct sums (use corollary 4.1.4), we have that $\hat{R} \otimes_R F \cong \hat{F}$, and so α is surjective, giving us the first part. If R is Noetherian, then N is finitely generated and so γ is surjective, and so we get exactness of the bottom line. Then by chasing the diagram we will get that α is injective, and hence an isomorphism, completing the proof.

From this, we get that when R is Noetherian, the extension of scalar functor $\widehat{(-)}$ is equivalent to the functor $\hat{R} \otimes_R (-)$ from the category of R -modules to the category of $\hat{R}$ -modules is exact. As a consequence:

Corollary 4.4.6: Flatness of Completion

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, $\hat{R}$ the $\mathfrak{a}$ -adic completion of R . Then $\hat{R}$ is a flat R -algebra.

Proof:

direct result of above discussion.

Proposition 4.4.7: Properties of Adic Completion

Let R be Noetherian and $\hat{R}$ its $\mathfrak{a}$ -adic completion. Then:

1. $\hat{\mathfrak{a}} = \hat{R}\mathfrak{a} = \hat{R} \otimes_R \mathfrak{a}$
2. $\widehat{\mathfrak{a}^n} = \hat{\mathfrak{a}}^n$
3. $\mathfrak{a}^n / \mathfrak{a}^{n+1} \cong \hat{\mathfrak{a}}^n / \hat{\mathfrak{a}}^{n+1}$
4. $\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{R}$

Proof:

exercise (or see Atiyah p.109)

We shall show the last one. By (2) and the fact that $\hat{\hat{R}} \cong \hat{R}$, we see that $\hat{R}$ is complete in the $\mathfrak{a}$ -adic topology. Hence for any $x \in \hat{\mathfrak{a}}$,

$$(1 - x)^{-1} = 1 + x + x^2 + \cdots$$

converges in $\hat{R}$, and so $1 - x$ is a unit. Then this implies that $\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{R}$, completing the proof.

Let us next characterize the kernel of the map $M \rightarrow \hat{M}$

Theorem 4.4.8: Krull's Theorem

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, M a finitely generated R -module, and $\hat{M}$ the $\mathfrak{a}$ -completion of M . Then the kernel

$$E = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$$

given by the map $M \rightarrow \hat{M}$ consist of all $x \in M$ that are annihilated by some element $1 + \mathfrak{a}$.

Proof:

It suffices to show that $(1 - \alpha)E = 0$. It is clear that if $(1 - \alpha)x = 0$

$$x = \alpha x = \alpha^2 x = \cdots \in \bigcap_{n=1}^{\infty} \mathfrak{a}^n M = E$$

Hence we shall show the converse. As $E \cap \mathfrak{a}^n M$, it is the intersection of all neighborhoods of $0 \in M$. Thus, in the induced topology, E is the only neighborhood of $0 \in E$, and hence the only neighborhood in the $\mathfrak{a}$ -adic topology. As $\mathfrak{a}E$ is a neighborhood in the $\mathfrak{a}$ -topology, we have that $\mathfrak{a}E = E$. As M is finitely generated and R is Noetherian, E finitely generated, and hence we have that:

$$\mathfrak{a}E = E \Rightarrow (1 - \alpha)E = 0$$

for some $\alpha \in \mathfrak{a}$, completing the proof.

Proposition 4.4.9: Localization is A Subring of Completion

Let R be Noetherian and let $S = 1 + \mathfrak{a}$. Then

$$S^{-1}R \hookrightarrow \hat{R}$$

Proof:

The map $R \rightarrow \hat{R}$ and $R \rightarrow S^{-1}R$ have the same kernel, and for any $\alpha \in \hat{\mathfrak{a}}$ we have

$$(1 - \alpha)^{-1} = 1 + \alpha + \alpha^2 + \cdots$$

converges in $\hat{A}$, and hence all S become a unit in $\hat{A}$. By the universal property of $S^{-1}A$, there is a natural homomorphism $S^{-1}A \rightarrow \hat{A}$ which is injective by theorem 4.4.8.

The Noetherian condition is necessary, an easy example lies within the realm of analysis:

Example 4.4.10: Localization and Completion Without Noetherian Property

Take R to be the collection of smooth functions on $\mathbb{R}$, $C^\infty(\mathbb{R})$. Let $\mathfrak{a}$ be the ideal of all f which vanish at 0. This ideal is maximal as $R/\mathfrak{a} \cong \mathbb{R}$. Then $\mathfrak{a}$ is generated by x (if you know some multi-variable calculus prove that any f that vanishes at 0 can be written as $f(x) = xg(x)$ for g smooth, it is a variation of Hadamard's lemma), and $E = \bigcap_{n=1}^{\infty} \mathfrak{a}^n$ is the set of all $f \in R$ such that all derivatives vanish at the origin. On the other hand, f is annihilated by some element $1 + \alpha$ for $\alpha \in \mathfrak{a}$ iff f vanishes identically in some neighborhood of 0. Now, the function e^{-1/x^2} has vanishing derivative at 0, but is not identically zero near 0. Hence the kernels of the $R \rightarrow \hat{R}$ and $R \rightarrow S^{-1}R$ with $S = 1 + \mathfrak{a}$ do not coincide.

This can also be a way of showing that R cannot be Noetherian.

Corollary 4.4.11: Krull Theorem, Noetherian Domain

Let R be a Noetherian domain and $(1) \neq \mathfrak{a} \subseteq R$ an ideal. Then

$$\bigcap \mathfrak{a}^n = 0$$

Proof:

Exercise; use the fact that $1 + \mathfrak{a}$ does not contain any zero-divisors.

Corollary 4.4.12: Krull Theorem, Hausdorff Topology

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal contained in the Jacobson radical, and M a finitely generated R -module. Then the $\mathfrak{a}$ -topology of M is Hausdorff, i.e.

$$\bigcap \mathfrak{a}^n M = 0$$

Proof:

By the Artin-Rees lemma, every element of $1 + \mathfrak{a}$ is a unit.

Corollary 4.4.13: Krull Theorem, Kernel Into Localization

Let R be a Noetherian ring, $\mathfrak{p} \subseteq R$ a prime ideal. Then the intersection of all $\mathfrak{p}$ -primary ideals of A is the kernel of $A \rightarrow A_{\mathfrak{p}}$

Proof:

The $\mathfrak{p}$ -primary ideals of A are the ideals contained in between $\mathfrak{p}$ and $\mathfrak{m}^n$. Corollary 4.4.12 implies that the intersection of all $\mathfrak{p}$ -primary ideals is 0. Then we use the correspondence between completions and localization by first lifting R from its localization and applying the map.

4.5 Associated Graded Rings: Noetherian Closed under Completion

The following will let us show another way of characterizing complete rings, and let us prove that the completion of a Noetherian ring is Noetherian.

Definition 4.5.1: Associated Graded Ring

Let R be a ring and $\mathfrak{a} \subseteq R$ an ideal. Then the *associated graded ring* is

$$G_{\mathfrak{a}}(R) = \bigoplus_{n=1}^{\infty} \mathfrak{a}^n / \mathfrak{a}^{n+1}$$

where $\mathfrak{a}^0 = A$. If unambiguous, $G(R)$ will be written instead of $G_{\mathfrak{a}}(R)$. For an R -module M with an $\mathfrak{a}$ -filtration (M_n) , the graded $G(R)$ -module is given by:

$$G(M) = \bigoplus_{n=0}^{\infty} M_n / M_{n+1} = \bigoplus_{n=0}^{\infty} G_n(M)$$

Multiplication is certainly well-defined: for each $x_n \in \mathfrak{a}^n$, $\bar{x}_n$ is the image of x_n in $\mathfrak{a}^n / \mathfrak{a}^{n+1}$. Then $\bar{x}_m \bar{x}_n$ is $\overline{x_m x_n}$ which is the image of $x_m x_n$ in $\mathfrak{a}^{m+n} / \mathfrak{a}^{m+n+1}$ (if the reader is not convinced, check to see if this is well-defined).

Proposition 4.5.2: Properties of Associated Graded Rings

Let R be a Noetherian ring and $\mathfrak{a} \subseteq R$ an ideal. Then:

1. $G_{\mathfrak{a}}(R)$ is Noetherian
2. $G_{\mathfrak{a}}(R)$ is isomorphic to $G_{\hat{\mathfrak{a}}}(\hat{R})$ as graded rings
3. If M is a finitely generated R -module and (M_n) is a stable $\mathfrak{a}$ -filtration of M , then $G(M)$ is a finitely generated graded $G_{\mathfrak{a}}(R)$ -module

Proof:

1. Use HBT in the right way
2. Use proposition 4.4.7(3)
3. exercise

Lemma 4.5.3: Associated Graded Ideal and Completion Map Relation

Let $\varphi : A \rightarrow B$ be a homomorphism of filtered groups ($\varphi(A_n) \subseteq B_n$), and let $G(\varphi) : G(A) \rightarrow G(B)$, $\hat{\varphi} : \hat{A} \rightarrow \hat{B}$ be the induced homomorphisms of the associated graded group and complete group respectively. Then:

1. $G(\varphi)$ is injective $\Rightarrow \hat{\varphi}$ is injective
2. $G(\varphi)$ is surjective $\Rightarrow \hat{\varphi}$ is surjective

Proof:

The key is to look at the following commutative diagram of exact sequences:

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & A_n/A_{n+1} & \longrightarrow & A/A_{n+1} & \longrightarrow & A/A_n & \longrightarrow & 0 \\
 & & \downarrow G_n(\varphi) & & \downarrow \alpha_{n+1} & & \downarrow \alpha_n & & \\
 0 & \longrightarrow & B_n/B_{n+1} & \longrightarrow & B/B_{n+1} & \longrightarrow & B/B_n & \longrightarrow & 0
 \end{array}$$

which by the snake lemma gives us the usual:

$$0 \rightarrow \ker G_n(\varphi) \rightarrow \ker \alpha_{n+1} \rightarrow \ker \alpha_n \rightarrow \operatorname{coker} G_n(\varphi) \rightarrow \operatorname{coker} \alpha_{n+1} \rightarrow \operatorname{coker} \alpha_n \rightarrow 0$$

Now, by induction on n , we see that:

1. $\ker \alpha_n = 0$
2. $\operatorname{coker} \alpha_n = 0$

taking the limit of the homomorphisms α_n and applying proposition 4.1.3 completes the proof.

Proposition 4.5.4: Partial Converse of lemma 4.5.3(3)

Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal of R , M an R -module, and (M_n) an $\mathfrak{a}$ -filtration of M . Suppose that R is complete in the $\mathfrak{a}$ -topology, that M is Hausdorff in its filtration topology so that $\bigcap_n M_n = 0$, and that $G(M)$ is a finitely generated $G(R)$ -module.

Then M is a finitely generated R -module.

Proof:

As $G(M)$ is finitely generated, pick some finite set of generators. Split them up into the homogeneous components, say ζ_i (with $1 \leq i \leq \nu$) where ζ_i has degree $n(i)$, and is the image of some $x_i \in M_{n(i)}$. Let F^i be the module with the stable $\mathfrak{a}$ -filtration given by $F_K^i = \mathfrak{a}^{k+n(i)}$ and let $F = \bigoplus_{i=1}^r F^i$. Then the mapping that takes the generator 1 of each F^i to x_i gives a homomorphism of filtered groups

$$\varphi : F \rightarrow M$$

Hence it induces a homomorphism of $G(A)$ -modules. By construction this map is surjective. Hence by lemma 4.5.3 $\hat{\varphi}$ is surjective. Now consider the diagram:

$$\begin{array}{ccc}
 F & \xrightarrow{\varphi} & M \\
 \alpha \downarrow & & \downarrow \beta \\
 \hat{F} & \xrightarrow{\hat{\varphi}} & \hat{M}
 \end{array}$$

As F is free and $A = \hat{A}$, α is an isomorphism. Next, as M is Hausdorff, β is injective. . As $\hat{\varphi}$ is surjective, φ is surjective, and so $x_1, \dots, x_r$ generate M as an A -module completing the proof.

Corollary 4.5.5: Partial Converse in Noetherian Case

Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal of R , M an R -module, and (M_n) an $\mathfrak{a}$ -filtration of M . Suppose that R is complete in the $\mathfrak{a}$ -topology, that M is Hausdorff in its filtration topology so that $\bigcap_n M_n = 0$, and that $G(M)$ is a finitely generated $G(R)$ -module.

Then if $G(M)$ is a Noetherian $G(R)$ -module, then M is a Noetherian R -module.

Proof:

If we show that every submodule $M' \subseteq M$ is finitely generated, we're done. Let $M'_n = M' \cap M_n$. Then (M'_n) is an $\mathfrak{a}$ -filtration of M' , and the embedding $M'_n \rightarrow M_n$ induces an injective homomorphism:

$$M'_n/M'_{n+1} \rightarrow M_n/M_{n+1}$$

Hence we get an embedding of $G(M')$ in $G(M)$, $G(M)$ is Noetherian, $G(M')$ is finitely generated, and so M' is Hausdorff. As $\bigcap_n M'_n \subseteq \bigcap_n M_n = 0$, we get that M' is finitely generated, as we sought to show.

Theorem 4.5.6: Noetherian Closed Under Completion

Let R be a Noetherian ring and $\mathfrak{a} \subseteq R$ an ideal. Then the $\mathfrak{a}$ -completion $\hat{R}$ of R is Noetherian

Proof:

As:

$$G_{\mathfrak{a}}(R) = G_{\hat{\mathfrak{a}}}(\hat{R})$$

is Noetherian, apply corollary 4.5.5 to the complete ring $\hat{R}$ and taking $M = \hat{R}$.

Corollary 4.5.7: Power Series Ring is Noetherian

Let R be a Noetherian ring, and let $S = R[[x_1, \dots, x_n]]$ be the power series ring in n variables. Then S is Noetherian.

Proof:

The ring $R[x_1, \dots, x_n]$ is Noetherian by Hilbert's Basis Theorem, and S is the completion for the $(x_1, \dots, x_n)$ -adic topology.

Ring Extension and Integral Extension

In this section, we generalize the notion of extension from fields to commutative rings. Just like how $k \subseteq F$ can be seen as a field extension, if we let k and R be rings instead, we will get the accompanying notion of ring extension:

Definition 5.0.1: Ring Extension

Let S be a commutative ring and $R \subseteq S$ a subring. Then we say that S is a *ring extension* of R . Equivalently, if there exists an injective ring homomorphism $\varphi : R \rightarrow S$, then S is a ring extension of R .

Unlike fields, we must specify the homomorphism to be injective, since ring homomorphisms need not be injective. The notation S/R is also used just as in field, though more scarcely. Just like how any field extension F/k makes F into a k -vector-space and we have special k -homomorphisms letting k -commute between the homomorphism, a ring extension S/R have an analogue: in fact, a k -homomorphism is *precisely* a k -algebra homomorphism!

Recall that any R -algebra can be defined as a ring homomorphism $\varphi : R \rightarrow S$, *provided* that $\varphi(R) \subseteq Z(S)$. Since we are working with commutative rings, this is *always* the case, and so we can always think of a ring extension S over R is an R -algebra! This is why we focus on *commutative rings*. There is a similar notion of ring extension for non-commutative which is closer to the notion of group extension (i.e. a short exact sequence extending the object), however as we have pointed out in remark ??, ring extensions don't break down into nice building blocks since kernels of rings are not rings, which means we would have to extend a ring R by an abelian group (i.e. two-sided ideal) I . This very much changes how we would study rings. However, we can always study the structure of any commutative ring S by how it behaves as an R -algebra similarly to how we studied fields, which

is what we'll do in these chapters.

5.0.2 Advanced Intuition Integral extensions shall correspond to *proper maps* of morphisms. Non-integral extensions shall never be proper, thus geometrically exhibiting greatly different behaviour. In particular, taking base change if necessary, the image of closed sets (essentially zero-sets of polynomials) need not be closed.

Perhaps the first notion we might want to generalize is the equivalent of an *algebraic element*, since they capture the relations within a ring extension with respect to the base ring. The following is the exactly this generalization:

Definition 5.0.3: Integral Elements

Let $R \subseteq S$.

1. An element $s \in S$ is called an *integral over R* if s is the root of a monic polynomial over $R[x]$:

$$s^n + r_{n-1}s^{n-1} + \cdots + r_1s + r_0 = 0 \quad r_i \in R$$

2. S is called *integral over R* or an *integral extension of R* if every element of S is integral over R . The integral closure of an integral domain is also called the *normalization of R* . Sometimes, the integral extension will be written as S/R .
3. The set $\overline{R}^S$ containing all the elements of S that are integral over R is called the *integral closure of R* .
4. If $R = \overline{R}^S$, then R is said to be *integrally closed over S* .
5. If R is an integral domain that is integrally closed in its field of fractions F , $\overline{R}^F$, then R is called *normal domain*.

The fact that the polynomial it satisfies has to be monic can be thought of as not wanting to add relations that produce “units” or unlit-like elements (ex. $3x - 1 = 0$ implies there is a new element that behaves like $1/3$). For a more specific motivation, we can think of the following: If $f(x) \in R[x]$ is a polynomial of minimal degree with root r , and $h(x) \in R[x]$ is another monic polynomial with minimal degree with root r , then we would want that $h(x)|f(x)$, just like for minimal polynomials over fields. In the process of doing so, we would require the coefficient of the leading terms to divide. Since they can be arbitrary, we would require the coefficients of these polynomials to be units, or just 1, in order for the division to follow through. It follows that it is a natural condition that all polynomials be monic for an integral extension.

Another motivation comes from Gauss’s lemma, which states that a polynomial $R[x]$ is irreducible iff it is irreducible in $k[x]$, granted that gcd of the coefficients of $p(x)$ is 1, or more simply that the leading coefficient is 1. This can be seen as well by the fact that field extension by linear equations (whether monic or not) are 1-dimensional equations, that is they don’t change the field! The parallel between factorization in $k[x]$ and $R[x]$ will become very important in chapter ??.

Integral elements share many of the same properties as algebraic elements:

Proposition 5.0.4: Integral Element Equivalences

Let $R \subseteq S$ be a subring of S . Then the following are equivalent:

1. s is integral over R
2. $R[s]$ is a finitely generated R -module
3. The element s is in T for some subring T where $R \subseteq T \subseteq S$, that is a finitely generated R -module

Proof:

$1 \Rightarrow 2$ Recall that $R[s]$ can be expressed as all R -linear combinations of powers of s . Since sd is integral over R :

$$s^n + r_n s^{n-1} + \cdots + r_1 s + r_0 = 0$$

in particular:

$$s^n = -(r_n s^{n-1} + \cdots + r_1 s + r_0)$$

It follows that for all $m \geq n$, s^m can be expressed in terms of $1, s_1, \dots, s_{n-1}$ (for the same reasons as we've shown for algebraic elements). Hence:

$$R[s] = R + Rs + \cdots + Rs^{n-1}$$

Notice here how if the polynomial was not monic, we might not have been able to isolate s^n , causing complications.

$2 \Rightarrow 3$ Let $T = R[s]$. Then the result immediately follows

$3 \Rightarrow 1$ We take advantage of rational canonical forms and Cramer's rule (see ref:HERE) (remember that if R is an integral domain, we can take its field of fractions, and the resulting matrix multiplication remains the same). Since T is finitely generated, let $v_1, v_2, \dots, v_n$ be a finite generating set. Since $s \in T$ and T is a ring, $sv_i \in T$ for all $1 \leq i \leq n$. Thus:

$$sv_i = \sum_j a_{ij} v_j$$

Giving us the equation n :

$$\sum_j (\delta_{ij} s - a_{ij}) v_j = 0 \quad 1 \leq i \leq n$$

where δ_{ij} is one when $i = j$ and 0 otherwise (i.e. the Kronecker delta). Letting B be the matrix such that the entries are $B_{ij} = \delta_{ij} s - a_{ij}$ and letting v be the $n \times 1$ matrix whose entries are $v_1, \dots, v_n$, we get:

$$Bv = 0$$

Thus, by Cramer's rule (ref:HERE):

$$(\det B) v_i = 0 \quad 1 \leq i \leq n$$

Now, since 1 is an R -linear combination of $v_1, v_2, \dots, v_n$,

$$\det B = (\det B)1 = (\det B) \sum_i a_i v_i = \sum_i a_i ((\det B) v_i) = 0$$

i.e. $\det B = 0$. Now, B is by definition $B = sI - A$ (where $A_{ij} = a_{ij}$). Thus, s is the root of a monic polynomial $\det(xI - A) \in R[x]$, namely the characteristic polynomial of A , and so is the root of a monic polynomial with coefficients in R , giving (1)

Corollary 5.0.5: Integral Elements Form Rings

Let $R \subseteq S$ be a subring of S , and $s, t \in S$. Then:

1. If s and t are integral over R , so is $s \pm t$ and st
2. $\overline{R}^S$ is a subring of S containing R , and is integrally closed^a

^aIf we were working over rng's, $\overline{R}^S$ need not be integrally closed

Proof:

Let s, t be integral over R , sy that by proposition 5.0.4 $R[s]$ and $R[t]$ are finitely generated R -modules, say:

$$\begin{aligned} R[s] &= Rs_1 + \dots + Rs_n \\ R[t] &= Rt_1 + \dots + Rt_m \end{aligned}$$

Then:

$$R[s, t] = Rs_1 t_1 + \dots + Rs_i t_j + \dots + Rs_n t_m$$

is a ring containing $s \pm t$ and st that is also a finitely generated R -module. It follows that The $\overline{R}^S$ is a subring which contains R , and is trivially integrally closed.

(note sure if I want the comment about "the process of taking integral closure stops after one step because integrality is transitive" that dummit puts on p. 693)

Note that integral closure is dependent on the ring over which we are closing; we do not have the equivalent notion of "integral closure of R " without specifying a ring over which it is integrally closed (Why?)

Corollary 5.0.6: Relating Finite Type And Finite Degree

Let S/R be an integral extension. Then S is a finitely generated R -module if and only if S is finite type over R and integral

Notice how if S is not integral, then it is possible that S is of finite type over R but not finitely generated over R , the most canonical example being $R[x]$ over R (since $R[x]$ is finitely generated as an R -algebra by 1 and x , but $1, x, x^2, \dots$ is an infinite set that is a basis for $R[x]$)

Proof:

If S is finitely generated as an R -module, it is certainly finite type over R , so we'll show the converse (which is a generalization of ??). Take Since S is of finite type, let's say $s_1, s_2, \dots, s_n$ generates S as a ring. Take

$$R \subseteq R[s_1] \subseteq R[s_1, s_2] \cdots \subseteq R[s_1, s_2, \dots, s_n]$$

Then each extension is a finitely generated field extension. Then (show as an exercise), $R[s_1, \dots, s_n]$ is finitely generated over R .

Before giving examples, we'll show that we have been studying special types of integral closures when we worked with field extensions in part ??:

Proposition 5.0.7: Integral Extension over Fields

Let S/R be an integral extension where S is an integral domain. Then R is a field if and only if S is a field

Note that if S was not an integral domain or commutative, this theorem does not work: Take $M_2(k)$ over any field. Then $M_2(k)$ is a 4-dimensional k -algebra, but certainly not a field (it contains zero-divisors, and is not commutative).

Proof:

($\Rightarrow$) Let R be a field and take $0 \neq s \in S$. Since s is integral over R , we have:

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0 = 0 \quad a_i \in R$$

Without loss of generality, let $a_0 \neq 0$; otherwise, factor an s , and using the fact that S is an integral domain, we see that either $s = 0$ or $s^{n-1} + \cdots + a_1 = 0$, and since $s \neq 0$, then we simply choose $s^{n-1} + \cdots + a_1 = 0$ to be our equation. Thus, moving a_0 and factoring s we get:

$$s(s^{n-1} + \cdots + a_1) = -a_0$$

Since R is a field, $\frac{-1}{a_0} \in R$, so:

$$s \cdot ((-1/a_0)(s^{n-1} + \cdots + a_1)) = 1$$

showing that s has a multiplicative inverse. Since s was arbitrary, we have that S is a field.

($\Leftarrow$) Let S be a field and take $0 \neq r \in R$ (automatically implying that R is an integral domain). Since $r \in R$, $r \in S$, and so $r^{-1} \in S$. Since S is an integral extension of R :

$$r^{-n} + a_{n-1}r^{-(n-1)} + \cdots + a_1r^{-1} + a_0 = 0$$

Multiply the entire expression by $r^{-(n+1)}$, and move all the terms except the r^{-1} terms to the right hand side to get:

$$r^{-1} = -(a_{n-1} + \cdots + a_1r^{n-2} + a_0r^{n-1})$$

which is a linear combination of elements in R , showing that $r^{-1} \in R$, and so R is a field.

Thus, any algebraic field extension is an integral extension.

Example 5.0.8: Integral Extensions

1. As we've just shown, all algebraic extensions of fields are integral extensions.
2. Let R be a Unique Factorization Domain and let $F = \text{Frac}(R)$. Then $\overline{R}^F = R$. The $\supseteq$ direction is obvious, so consider the $\subseteq$ direction: suppose that $a/b \in \overline{R}^F$ with $\gcd(a, b) = 0$. Then by definition

$$\left(\frac{a}{b}\right)^n + r_{n-1} + \left(\frac{a}{b}\right)^{n-1} + \cdots + r_1 \frac{a}{b} + r_0 = 0$$

Then:

$$a^n = b(-r_{n-1}a^{n-1} - \cdots - r_1ab^{n-2} - r_0b^{n-1})$$

so $b|a^n$, and hence $b|a$. Since $\gcd(a, b) = 1$, it must thus be that b is a unit, and hence we have that $a/b \in R$, showing that $\overline{R}^F = R$. Hence, all UFD's are normal domains. This should make intuitive sense, since any element $a/b \in \text{Frac}(R)$ would satisfy the equation $bx - a = 0$, which is not-monic.

More generally, any polynomial ring over a UFD R , $R[x_1, \dots, x_n]$ is integrally closed over its field of fractions $\text{Frac}(R[x_1, \dots, x_n])$ (by ??), showing every polynomial ring over a Unique Factorization Domain is a normal domain.

3. Let S be integral over R and $I \subseteq S$ be an ideal. Then S/I is integral over $R/(R \cap I)$. To see this, notice that a monic polynomial over R where $s \in S$ is a root, when reduced modulo I gives a monic polynomial over $R/(R \cap I)$ where $\bar{s} \in S/I$ is a root.
4. Let k be any field and take $k[x, y]$ (which is integrally closed since it's a UFD). Take the prime ideal $(x^2 - y^3) \subseteq k[x, y]$ (see exercise ref:HERE, 14 in section 9.1 in Dummit and Foote). Consider the integral domain

$$R = k[x, y]/(x^2 - y^3) = k[\bar{x}, \bar{y}]$$

Notice that R is *not* normal: the element $\bar{x}/\bar{y}$ is the solution to the equation

$$\left(\frac{\bar{x}}{\bar{y}}\right)^3 - \bar{x} = 0$$

but evidently $\bar{x}/\bar{y} \notin R$. Thus, R cannot be a Unique Factorization Domain.

If you wish to practice, another example is $\mathbb{Z}[\sqrt{-3}]$. Its integral closure over its field of fractions is:

$$\mathbb{Z}\left[\frac{1 + \sqrt{-3}}{2}\right]$$

5.1 Extension and Contraction of Ideals

Probably the biggest difference between fields and rings is that ideals are nontrivial. Thus, it is meaningful to ask what happens to ideals in (general) ring extensions. Recall that if $f : R \rightarrow S$ is a

ring homomorphism, and $J \subseteq S$ is an ideal, $f^{-1}(S)$ is an ideal (re-prove this if you're not convinced). However, if $I \subseteq R$ is an ideal, $f(I)$ need not be an ideal (take $\mathbb{Z} \hookrightarrow \mathbb{Q}$ with any non-trivial ideal of $\mathbb{Z}$). The next best thing would be to define the smallest ideal of B containing $f(I)$:

Definition 5.1.1: Extension and Contraction

Let $\varphi : R \rightarrow S$ be a homomorphism between commutative rings and let $I \subseteq R$, $J \subseteq S$.

1. The *extension* of I to S is the ideal $\varphi(I)S$, commonly denoted I^e
2. The *contraction* of J in R is the ideal $\varphi^{-1}(J)$, commonly denoted I^c

In the case where φ is the natural inclusion $\iota : R \rightarrow S$, then the extension of I is IS , and the contraction of J is $J \cap R$. If it is clear that $I \subseteq R$, there are multiple extension of R , say S/R and T/R , then the extension of I is sometimes denoted I_S or I_T to emphasize to what ring I is being extended too S or T respectively.

Proposition 5.1.2: Properties of Extensions and Contractions

1. $I \subseteq IS \cap R$, or more generally that I is contained in the contraction of its extension to S ($I \subseteq I^{ec}$)
2. $(J \cap R)S \subseteq J$, or more generally that J is contained in the extension of its contraction in R ($J \supseteq J^{ce}$)
3. $I^e = I^{ece}$
4. $J^c = J^{cec}$
5. If J is prime, J^c is prime. If I is prime, I^e need not be prime (consider $\mathbb{Z} \hookrightarrow \mathbb{Q}$). If $\varphi : R \rightarrow S$ is surjective, then I^e is prime.
6. If J is maximal, J^c need not be maximal, and similarly if I is maximal then I^e need not be maximal.

Proof:
exercise

Lemma 5.1.3: Extension And Contraction With Common Operations

Let $\mathfrak{a}_1, \mathfrak{a}_2 \subseteq A$ be ideals of A and $\mathfrak{b}_1, \mathfrak{b}_2 \subseteq B$ be ideals of B . Then:

$$\begin{array}{ll}
 (\mathfrak{a}_1 + \mathfrak{a}_2)^e = \mathfrak{a}_1^e + \mathfrak{a}_2^e & (\mathfrak{b}_1 + \mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c + \mathfrak{b}_2^c \\
 (\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e & (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c = \mathfrak{b}_1^c \cap \mathfrak{b}_2^c \\
 (\mathfrak{a}_1 \mathfrak{a}_2)^e = \mathfrak{a}_1^e \mathfrak{a}_2^e & (\mathfrak{b}_1 \mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c \mathfrak{b}_2^c \\
 (\mathfrak{a}_1 : \mathfrak{a}_2)^e = (\mathfrak{a}_1^e : \mathfrak{a}_2^e) & (\mathfrak{b}_1 : \mathfrak{b}_2)^c \subseteq (\mathfrak{b}_1^c : \mathfrak{b}_2^c) \\
 \sqrt{\mathfrak{a}}^e \subseteq \sqrt{\mathfrak{a}^e} & \sqrt{\mathfrak{b}}^c = \sqrt{\mathfrak{b}^c}
 \end{array}$$

Proof:
exercise.

If $\varphi : R \rightarrow S$ is a ring homomorphism, we may decompose φ into the following:

$$R \xrightarrow{p} \varphi(R) \xrightarrow{j} S$$

where p maps R surjectively onto $\varphi(R)$, and j injects into S . Relating the ideals of R to the ideals of $\varphi(R)$ is very easy:

Proposition 5.1.4: Correspondence Between Ideals And Image Of Ideals

Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then there exists a bijective correspondence:

$$\{I \subseteq \varphi(R)\} \xrightleftharpoons[\varphi^c]{\varphi^e} \{\ker(\varphi) \subseteq J \subseteq R\}$$

Furthermore, prime ideals of $\varphi(R)$ correspond to prime ideals of R .

Proof:
exercise

On the otherhand, the correspondence of $\varphi(R)$ and S is much harder, and shall be covered in section 5.2 when S/R is an integral extension. As shown in proposition 5.1.2, every prime ideal “lies over” another prime ideal. This motivates the following vocabulary:

Definition 5.1.5: Lying Over

Let $I \subseteq S$ be an ideal and S/R an integral extension. If $J = I \cap R$, I is said to be *lying over* J .

Example 5.1.6: Extending ideals

Take $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}[i]$ to be the inclusion. Then a prime ideal $(p) \subseteq \mathbb{Z}$ may or may not remain prime when extended to $\mathbb{Z}[i]$. Using Fermat’s theorem of sums of squares (??), it can be deduced that:

1. $(2)^e = (1 + i)^2$
2. If $p \equiv 1 \pmod{4}$, then p^e is the product of two distinct prime ideals (ex. $(5)^e = (2 + i)(2 - i)$)
3. if $p \equiv 3 \pmod{4}$, then p^e is prime in $\mathbb{Z}[i]$

In general, it is rather hard to determine when $\varphi(p)$ is prime when p is prime. If $\varphi : R \rightarrow S$ is an integral extension, then the situation becomes much better, however it will require the notion of *localization* which shall be covered in the next section. One special case is note mentioning:

Proposition 5.1.7: Maximal Ideals Lying Over

Let S/R and $\mathfrak{m} \subseteq S$ be an ideal. Then $\mathfrak{m} \cap R$ is maximal if and only if $\mathfrak{m}$ is maximal

Proof:

Consider the integral domain $S/\mathfrak{m}$. Then $S/\mathfrak{m}$ is an integral extension of $R/(\mathfrak{m} \cap R)$. Then by proposition 5.0.7, $S/\mathfrak{m}$ is a field if and only if $R/(\mathfrak{m} \cap R)$ is a field. But then $\mathfrak{m}$ is maximal if and only if $\mathfrak{m} \cap R$ is maximal.

5.2 Integral Extensions and Prime Ideals

We return to the question of the extension of ideals, in particular prime ideals, given S/R :

Theorem 5.2.1: Integral Extension Prime Correspondence (Cohen–Seidenberg Theorem)

Let S be an integral extension of the integral domain R

1. **(Lying-over Property)** Let $\mathfrak{p} \subseteq R$ be a prime ideal. Then there is a prime ideal $P \subseteq S$ such that $\mathfrak{p} = P \cap R$. Furthermore, $\mathfrak{p}$ is maximal if and only if P is maximal.
2. **(Going-up Property)** Let $p_1 \subseteq p_2 \subseteq \cdots \subseteq p_n$ be a chain of prime ideals of R , and suppose there exists a chain $q_1 \subseteq q_2 \subseteq \cdots \subseteq q_m$, ($m < n$) such that $p_i = q_i \cap R$. Then there exists prime ideals $q_{m+1}, \dots, q_n$ that complete the chain: $q_m \subseteq q_{m+1} \subseteq \cdots \subseteq q_n$, $p_i = q_i \cap R$ for all i .
3. **(Going-down Property)** If R is a normal domain, $p_1 \supseteq p_2 \supseteq \cdots \supseteq p_n$ be a chain of prime ideals of R , and suppose there exists a chain $q_1 \supseteq q_2 \supseteq \cdots \supseteq q_m$, ($m < n$) such that $p_i = q_i \cap R$. Then there exists prime ideals $q_{m+1}, \dots, q_n$ that complete the chain: $q_m \supseteq q_{m+1} \supseteq \cdots \supseteq q_n$, $p_i = q_i \cap R$ for all i .
4. **(Incomparability property)** Let q_1 and q_2 lie over p . Then either $q_1 = q_2$, or they do not contain each other: $q_1 \not\subseteq q_2$ and $q_2 \not\subseteq q_1$

Proof:

1. We'll take advantage of the correspondence theorem for rings and their localization. Let $D = R - P$. Then as we saw, D is multiplicatively closed, and so $R[D^{-1}]$ is well-defined. Since D is also a multiplicatively closed set of S , $S[D^{-1}]$ is also well-defined. Furthermore, $S[D^{-1}]/R[D^{-1}]$ is an integral extensions of $R[D^{-1}]$: for any $s \in S[D^{-1}]$, if $s \in S$, clearly S is still integral. If $t = \frac{s}{d}$ for some $s \in S$ and $d \in D$, then letting

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0 = 0$$

This equation is still zero if we multiply by d^{-n} :

$$\frac{s^n}{d^n} + \frac{a_{n-1}s^{n-1}}{d^n} + \cdots + \frac{a_1s}{d^n} + \frac{a_0}{d^n} = 0$$

and shuffling the coefficients we get:

$$\frac{s^n}{d^n} + \frac{da_{n-1}s^{n-1}}{d^{n-1}} + \cdots + \frac{d^{n-1}a_1s}{d} + \frac{a_0}{d^n} = 0$$

and replacing each $d^{n-i}a_i = b_i$, and $s/d = t$,

$$t^n + b_{n-1}t^{n-1} + \cdots + b_1t + b_0 = 0$$

showing that t is integral over $R[D^{-1}]$, and so $S[D^{-1}]$ is an integral extension of $R[D^{-1}]$. Let $m \subseteq S[D^{-1}]$ be a maximal ideal. Then by the correspondence of ideals from a ring and its localization, we have that $m \cap R_p$ is a maximal ideal in R_p , and $m \cap R_p$ is the extension of P to the local ring R_p , so it contracts to p . It might be easier to see if we have a visual:

$$\begin{array}{ccc} S & \xrightarrow{\pi} & S[D^{-1}] \\ \uparrow \iota & & \uparrow \iota \\ R & \xrightarrow{\pi} & R[D^{-1}] = R_p \end{array}$$

The maximal ideal m is maximal in $S[D^{-1}]$, its pre-image is a maximal ideal of $R[D^{-1}]$, and the pre-image of that map is p , showing that $p = m \cap R$, as we sought to show.

2. This can be reduced to showing that if $p_1 \subseteq p_2$ and q_1 lies above p_1 such that $p_1 = q_1 \cap R$, then there exists a q_2 that lies above p_2 ($q_2 = p_2 \cap R$) and $q_1 \subseteq q_2$. First, let $\bar{S} = S/q_1$ and $\bar{R} = R/p_1$. Then $\bar{S}/\bar{R}$ is an integral extension by similar reasoning to what we've shown in part 1. By the first question in this homework, we have that p_2/p_1 is a prime ideal of $\bar{R}$. Then by part (1), we have that there exists an ideal $q_2/q_1 \subseteq \bar{S}$ that lies above p_2/p_1 , in particular:

$$q_2/q_1 \cap \bar{R} = p_2/p_1$$

At this point we are essentially done: taking the pre-image of q_2/q_1 , we get q_2 , and by the correspondence theorem we get

$$q_2 \cap R = p_2$$

3. here
4. For the sake of contradiction, let $q_1 \subseteq q_2$, and let both lie over p : $q_1 \cap R = q_2 \cap R = p$. Consider $\bar{S} = S/q_1$ and $\bar{R} = R/p$. Then as we've shown earlier, $\bar{S}/\bar{R}$ is an integral extension, and $q_2/q_1 \subseteq \bar{S}$ is a prime ideal. Consider:

$$(q_2/q_1) \cap \bar{R} = (q_2 \cap R)/(R \cap q_1) = (R \cap q_1)/(R \cap q_1) = \{0\}$$

Thus, $q_2/q_1 = (\{0\}) \subseteq \bar{S}$, and so $q_2 \subseteq q_1$, so $q_2 \subseteq q_1 \subseteq q_2$, showing that they are in fact equal.

Exercise 5.2.1

1. Let $I \subseteq R$ be an ideal and R a ring. Then if we consider integral extension of *rngs*, show that I^{n+1}/I^n is an integral extension.
2. Let $I \subseteq R$, $f : R \rightarrow S$ an integral extension and $I^e \subseteq S$ the extension of S . Then:

$$S/I^e \cong S \otimes_R R/I$$

6

Valuation Ring and Local Rings

Recall from chapter ??, section ?? that the absolute value defines a notion of size on any field. If the field is finite, the absolute is trivial and hence doesn't define any notion of size satisfying the axioms of absolute value besides $|x| = 1$ if $x \neq 0$. By ??, up to isomorphism the only absolute values on $\mathbb{Q}$ are $|\cdot|_p$ and $|\cdot|$. These absolute values were defined using a constant c^{-m} , however the choice of this constant did not affect the topology, and hence what really matters was the exponent. In this section, we define a function that captures the exponent information. It shall be a useful concept as a link between geometric and algebraic intuition's on certain fields.

Definition 6.0.1: $\mathbb{R}$ -Valuation

Let K be a field. Append ∞ to $\mathbb{R}$ and preserve the total ordering by adding the properties that $g \leq \infty$ for all $g \in \mathbb{R}$ and $g + \infty = \infty + g = \infty + \infty = \infty$. Then a function $\nu : K \rightarrow \mathbb{R} \cup \{\infty\}$ is called a $\mathbb{R}$ -valuation if

1. $\nu(K^\times) \subseteq \mathbb{R}$ and $\nu|_{K^\times} : K^\times \rightarrow \mathbb{R}$ is a group-homomorphism, that is $\nu(x \cdot y) = \nu(x) + \nu(y)$.
2. $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$
3. $\nu(x) = \infty$ if and only if $x = 0$

It may be more clear that $\mathbb{R}$ -valuations come as a generalization of non-Archimedean absolute values from the original definition given by Emil Artin (and hence why these are called valuations). To make sense of this definition, we shall introduce the slightly more general notion of a totally ordered abelian group.

Definition 6.0.2: Totally Ordered [Abelian] Group

Let G be an abelian group. Then G is said to be a *totally ordered group*, or simply an *ordered group* if $\leq$ is a total ordering on G that is translation invariant, that is for all $x, y, z \in G$ where $x \leq y$:

$$zx \leq zy$$

It was shown by ref:HERE¹ that an abelian group is totally order iff it is torsion-free. Not all torsion-free abelian groups can be extended to a totally ordered field k , and hence they cannot fit into $\mathbb{Q} \subseteq k \subseteq \mathbb{R}$ (see ??). Hence, this is a proper generalization, but not one that will not be needed beyond connecting $\mathbb{R}$ -valuations to absolute values defined in chapter ??:

Example 6.0.3: Valuation Ring

1. Puiseau series has valuation group $\mathbb{Q}$. See Richard Borchard Schemes 22: Valuation rings

Now, in definition 6.0.1, the $(\mathbb{R}, +)$ can be replaced with $(G, *)$. The following shows how this links to absolute values:

Definition 6.0.4: G -Valuation Equivalent Definition

Let K be a field. Append 0 to G and preserve the total ordering by adding the properties that $0 \leq g$ for all $g \in G$ and $0 * g = g * 0 = 0 * 0 = 0$. Then a function $\nu : K \rightarrow G \cup \{0\}$ is called a *G -valuation* if

1. $\nu(K^\times) \subseteq G$ and $\nu|_{K^\times} : K^\times \rightarrow G$ is a group-homomorphism, that is $\nu(x \cdot y) = \nu(x) * \nu(y)$.
2. $\nu(x + y) \leq \max\{\nu(x), \nu(y)\}$
3. $\nu(x) = 0$ if and only if $x = 0$

The image of $\nu(K^\times)$ is called the *value group*.

If $G = \mathbb{Z}$, or equivalently in the $\mathbb{R}$ -valuation definition, when $v(K^*) \subseteq \mathbb{Z}$, a special name is given:

Definition 6.0.5: Discrete Valuation

Let K be a field. Then a function $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ is called a *discrete valuation* if

1. $\nu(x \cdot y) = \nu(x) + \nu(y)$
2. $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$
3. $\nu(x) = \infty$ if and only if $x = 0$

Example 6.0.6: Valuations

1. Given any non-Archimedean absolute value v_p with values in $(\mathbb{R}^+, \cdot)$, we get a $\mathbb{R}^+$ valuation. Hence, each v_p is a valuation on $\mathbb{Q}$ in the equivalent definition, and v_∞, v_0 are valuations on

¹got it from [this](#) which lead to the comment here:

$k(x)$ for any field k (what has to be modified for the modern definition?).

Given a field k , it may be asked what valuations may be put on it. This is in correspondence with certain sub-rings of k :

Definition 6.0.7: Valuation Ring

Let ν be a valuation on K . Then

$$\mathcal{O}_K = \{x \in K \mid \nu(x) \geq 0\} \quad (\text{equiv. } \{x \in K \mid \nu(x) \leq 1\})$$

forms a ring, called the *Valuation Ring*. If ν is a discrete valuation, the ring is called a *discrete valuation ring*.

The reader should verify this is indeed a ring.

Proposition 6.0.8: Valuation Ring Properties

Let k be a field with valuation ν and let $\mathcal{O}_k$ be the valuation ring associated to it. Then:

1. Given $x \in k \setminus \{0\}$, either $x \in \mathcal{O}_k$ or $x^{-1} \in \mathcal{O}_k$ (where both are possible at once). As a consequence, k is a field of fractions of $\mathcal{O}_k$.^a
2. The group of units is $\mathcal{O}_K^* = \{x \in k \mid \nu(x) = 0\}$ (equiv. $\nu(x) = 1$)
3. $\mathcal{O}_k$ has exactly one maximal ideal $\mathfrak{m}_v = \{x \in k \mid \nu(x) > 0\} = \mathcal{O}_k \setminus \mathcal{O}_k^*$ (equiv $\nu(x) < 1$)
4. The ideal of $\mathcal{O}_k$ form a total ordering under $\subseteq$
5. The ring $\mathcal{O}_k$ is integrally closed (i.e. a normal domain)

^aNote that the either-or implies a general Γ -valuation!

Proof:

Points 1-3 are left as an exercise.

For 4, first show it on principal ideals, namely if $(a), (b) \subseteq \mathcal{O}_K$, then $(a) \subseteq (b)$ or $(b) \subseteq (a)$. Do this by noticing that either $ab^{-1} \in \mathcal{O}_K$ or $a^{-1}b \in \mathcal{O}_K$. For general ideals, if $\mathfrak{a}$ and $\mathfrak{b}$ are ideals that are not ordered, then there exists $a \in \mathfrak{a} \setminus \mathfrak{b}$ and $b \in \mathfrak{b} \setminus \mathfrak{a}$. Consider $(a), (b)$ which must be ordered namely $(a) \subseteq (b)$ or $(b) \subseteq (a)$. Say $(a) \subseteq (b)$. Then $(a) \cap (b) = (a)$. But then $a \in (b)$ meaning $a = bx$ for some $x \in \mathcal{O}_K$. But as $bx \in \mathfrak{b}$, this would imply $a \notin \mathfrak{a} \setminus \mathfrak{b}$, a contradiction.

We shall show that $\mathcal{O}_K$ is integrally close. Say that $x \in K$ is integral over $\mathcal{O}_K$, for the sake of contradiction say $x \notin \mathcal{O}_K$. As it is integral:

$$x^n + a_1x^{n-1} + \cdots + a_n = 0$$

with $a_i \in \mathcal{O}_K$. As $x \notin \mathcal{O}_K$, $x^{-1} \in \mathcal{O}_K$. But then:

$$x = -a_1 - a_2x^{-1} - \cdots - a_n = 0$$

But then $x \in \mathcal{O}_K$, a contradiction, completing the proof.

These properties in fact characterize valuation rings:

Proposition 6.0.9: Characterizing Valuation Ring

Let k be a field, $R \subseteq k$ a sub-ring, and R^* the group of units of R . Then the following are equivalent:

- ($\Rightarrow$) R is a valuation ring of k
- ($\Rightarrow$) $k = \text{Frac}(R)$ and the ideals form a chain:
- ($\Rightarrow$) for $x \in k \setminus \{0\}$, either $x \in R$ or $x^{-1} \in R$

Then $G = (k \setminus \{0\})/R^*$ is a totally ordered abelian group and

$$\nu_R : x \mapsto xR^*$$

is a valuation on k . The induced valuation ring is R . As a consequence, two valuations on the same field if and only if they have the same valuation ring

Proof:
exercise.

6.0.1 Discrete Valuation Ring

Limiting to discrete valuation rings, then there is a smallest positive value s such that:

$$\nu(K^*) = s\mathbb{Z}$$

We can normalize this result and get an equivalent ν where $\nu(K^*) = \mathbb{Z}$ and $\mathcal{O}_K$, $\mathcal{O}_K^*$, and $\mathfrak{p}$ are the same. Then there must be an element such that

$$\nu(\pi) = 1$$

Labelling this element as π , we can use it to show that any $x \in K \setminus \{0\}$ can be written as :

$$x = u\pi^m$$

for a unit $u \in K^\times$ ($\nu(u) = 0$). To see this, let

$$\nu(x) = m \Rightarrow \nu(x\pi^{-m}) = 0 \Rightarrow u = x\pi^{-m} \in \mathcal{O}_K^*$$

which gives the result. This works as $m \in \mathbb{N}$. This in fact characterizes DVR's:

Proposition 6.0.10: Characterizing DVR

Let R be an integral domain and $k = \text{Frac}(R)$. Then R is the valuation ring of a discrete valuation ν on k if and only if R is a PID with a unique nonzero prime ideal.

Proof:
exercise

Example 6.0.11: Discrete Valuation Ring

1. The ring of p -adic integers $\mathbb{Z}_p$ is a discrete valuation ring.
2. For any prime p , $\mathbb{Z}_{(p)}$ the localization of $\mathbb{Z}$ at the prime (p) is a discrete valuation ring.
3. The ring $k[[x]]$ of formal power series is a DVR with unique non-zero prime ideal (x) .
4. (If you know some analysis) Take the ring of Real or complex power series that converge to 0. Show this is a DVR.
5. (If you know some complex analysis) Let X be a Riemann surface. and let $M(X)$ be the collection of meromorphic functions $X \rightarrow \mathbb{C} \cup \{\infty\}$. Then $M(X)$ is a field. For each point $p \in X$, define a discrete valuation on $M(X)$ where

$$\nu_p(f) = i$$

where i is the largest number st

$$\frac{f(z)}{(z-p)^i}$$

can be extended to a holomorphic function (i.e. the order of the root/pole).

It shall in algebraic geometry that DVR's appear naturally in the setting of "smooth projective algebraic curves".

Proposition 6.0.12: Representation Of Elements Of DVR

Let k be a field with DVR $\mathcal{O}_k$. Let $\mathcal{T} \subseteq \mathcal{O}_k$ where $0 \in \mathcal{T}$ and one element of every coset of $\mathfrak{m}$ is in $\mathcal{T}$. Then every element $x \in \mathcal{O}_k$ is the sum of a unique power series:

$$\sum_{i \geq 0} r_i p^i$$

with coefficients $r_i \in \mathcal{T}$. Every nonzero $x \in k$ is the sum of unique laurent series $\sum_{i \geq m} r_i p^i$ with coefficients $r_i \in \mathcal{T}$ for all $i \geq m$ and $r_m \neq 0$

Proof:
exercise.

Exercise 6.0.1

1. Show the following properties (or examples) of ordered groups (ex. torsion-free)
2. Let $\mathcal{O}$ be a DVR with unique prime $\mathfrak{p}$. Show that $\mathfrak{p}^n / \mathfrak{p}^{n+1} \cong \mathcal{O} / \mathfrak{p}$ for $n \geq 0$

3. Let $x \in R$ for a valuating ring R . Then $1 - x$ is a unit. To make this intuitive, consider the fact that $1/(1 - x)$ always has a power-series expansion (see ??)

Artinian Rings

As we've seen, being Noetherian has many powerful consequences by allowing to bring in a consistent notion of finiteness into rings, lending itself useful in being able to define concepts like dimension. It is thus a natural question to ask if the dual condition can give us any interesting results, namely the *descending chain condition*. In this section, we explore the properties of rings and commutative rings satisfying the descending chain condition. It turns out that the D.C.C. is much more restrictive,

(The place they seem to be important is that a simple ring that is also Artinian is, by Artin-Wedderburn, isomorphic to a product of matrix ring over divisor rings)

Definition 7.0.1: Artinian module and ring

Let M be a R -module. Then if M satisfies the *descending chain condition* on submodules, it is an *artinian module*. More precisely, for any collection of submodules such that

$$M_1 \supseteq M_2 \supseteq \cdots \supseteq M_k \supseteq$$

then for some n , we get $M_n = M_{n+1} = \cdots$.

If we let $M = R$ be a module over itself, then we say that R is an *Artinian ring*.

Naturally, an Artinian ring has the D.C.C. condition on its ideals.

Proposition 7.0.2: Artinian Module Equivalent

M is Artinian if and only if every non-empty set of submodules of M contains a minimal element under inclusion

Proof:

Same proof as for the Noetherian case, just reversed

Just like Noetherian modules, if $N \subseteq M$ and M/N are artinian, so is M .

Lemma 7.0.3: Artinian Module Closed Under Quotient

Let M be an Artinian module and $N \subseteq M$. Then M/N is also Artinian

Proof:

exercise

Example 7.0.4: Artinian, But Not Noetherian Module

Take $\mathbb{Z}[1/2]/\mathbb{Z}$ over $\mathbb{Z}$. This ring is in the appropriate sense isomorphic to:

$$\mathbb{Z}/\mathbb{Z} \subseteq \mathbb{Z}/2\mathbb{Z} \subseteq \mathbb{Z}/4\mathbb{Z} \subseteq \cdots$$

which we see is an infinite increasing sequence. With a bit of work, we can show these are the only submodules, and hence we have an Artinian, but not Noetherian ring.

However, unlike Noetherian rings, Artinian rings need to satisfy stronger properties on their ideals:

Proposition 7.0.5: Artinian, Prime $\Rightarrow$ Maximal

Let R be an Artinian ring and $P \subseteq R$ be a prime ideal. Then P is maximal

Notice how this differs from Noetherian rings: $\mathbb{Z}[x]$ is Noetherian, (x) is prime, but not maximal since $(x) \subsetneq (2, x)$. Thus, being Artinian immediately imposes stronger conditions on its ideals.

Proof:

Consider R/P , which is also artinian by lemma 7.0.3. We'll show $R/P = F$ is a field. Consider $x \in F \setminus \{0\}$. Then by the descendign chain condition:

$$(x) \supseteq (x^2) \supseteq \cdots$$

must staxalize, that is for some n :

$$(x^n) = (x^{n+1})$$

which implies $x^n = x^{n+1}c$. By the Cancellion law, we get $1 = xc$, showing that x is in fact a unit. Since every element is a unit in F , we have that F is in fact a field, and so P is maximal, as we sought to show.

(This seems out of place; will move into comm alg somewhere)

Definition 7.0.6: Krull Dimension

Let R be a commutative ring. Then the *krull dimension* of R (or simply the *dimension* of R) is the longest possible length of distinct prime ideals (i.e. the supremum of the length of chains of [distinct] prime ideals):

$$P_1 \subsetneq P_2 \subsetneq \cdots \subsetneq P_n \subsetneq \cdots$$

where we count the number of inclusions $\subsetneq$. If R admits arbitrarily long chains of distinct prime ideals, then R is said to be infinite dimensional. The Krull dimension of a ring is often denoted $\text{kdim}(R)$ or $\dim(R)$.

(mention how krull dimension matches the transcendence degree if R is a finitely generated k -algebra. The advantage of this definition is that it does not refer to anything about finitely generated k -algebra, just arbitrary commutative rings)

Example 7.0.7: Krull Dimension

1. If $R = F$ is a field, then $\dim(F) = 0$
2. if R is a principal ideal domain, then since all prime ideals are maximal, $\dim(R) = 1$.
3. We will prove later that if k is a field, then $k[x_1, \dots, x_n]$ has Krull dimension n .
4. We will prove later that if R is a noetherian ring with dimension n , then $\dim(R[x]) = n + 1$
5. We will soon show that any artinian ring has dimension 0

Theorem 7.0.8: Hopkin's Theorem

Let R be an Artinian ring. Then:

1. There are only finitely many maximal ideals
2. The quotient $R/(\text{Jac}(R))$ is the direct product of a finite number of fields, where if there are n maximal ideals, we have a direct product of n fields R/M_i .
3. Every prime ideal of R is maximal. Thus $\text{kdim}(R) = 0$. As a consequence, $\text{Jac}(R) = \text{Nil}(R)$ and is a nilpotent ideal, $(\text{Jac}(R))^n = 0$ for some $n \geq 1$
4. R is isomorphic to a direct product of a finite number of local artinian rings
5. Every Artinian ring is Noetherian

Note that the last conditions is not true for Artinian Modules.

Proof:

1. Consider the chain:

$$m_1 \supseteq m_1 \cap m_2 \supseteq \cdots$$

I first claim that each inclusion of this chain is proper. To see this, recall that if m_i and

m_j are non-equal maximal ideal, it must be that $m_i + m_j = 1$. Thus, any collection of maximal ideals has this pair-wise condition. By the Chinese Remainder Theorem:

$$R/(m_1 \cap \cdots \cap m_n) \cong R/m_1 \times R/m_2 \times \cdots \times R/m_n$$

Now simply see that the element $(\bar{0}, \dots, \bar{0}, \bar{1})$ is non-zero, and hence not contain in $m_1 \cap \cdots \cap m_n$, showing that each inclusion *must* be proper! Thus, by the D.C.C., the above chain must stabalize, meaning there can only be finitely many maximal ideals.

2. The proof follows immediately from the previous observation of the CRT
3. We've shown $\text{Jac}(R)$ is nilpotent, hence $\text{Jac}(R) \subseteq \sqrt{(0)}$. Since every prime ideal contains the nilradical (recall if $0 = x^n \in P$, then either $x \in P$ or $x^{n-1} \in P$, which inductively shows that $x \in P$), hence contain $\text{Jac}(R)$. Thus, considering the natural projection $\pi : R \rightarrow R/\text{Jac}(R)$, it is clear that $\pi(P)$ is also prime. Since

$$R/\text{Jac}(R) \cong k_1 \times \cdots \times k_n$$

and since ideals in product rings $R_1 \times R_2$ are of the form $I_1 \times I_2$, we see that the image being prime implies it must be a subset of the codomain where either the component is all of k_i , or 0. It follow's that the image if a maximal ideal, and so by the correspondence theorem P must be a maximal ideal in R , showing that all prime ideals are maximal

4. Let $m_1, \dots, m_n$ be the maximal ideals of R and let n be the number such that $(\text{Jac}(R))^k = 0$. Then

$$\prod_i^n M_i^k \subseteq \left(\prod_i^n M_i \right)^k \subseteq (\text{Jac}(R))^k = 0$$

Thus, by the CRT

$$R \cong (R/M_1^k) \times \cdots \times (R/M_n^k)$$

Notice that each R/M_i^k is an artinian with unique maximal ideal M_i/M_i^k .

5. By part (4), it suffices to show that local Artinian rings are Noetherian. If m is the maximal ideal of the local artinian ring R , then $m = \text{Jac}(R)$, so $m^k = (\text{Jac}(R))^k = 0$ for some $k \in \mathbb{N}_{>0}$. Hence $R \cong R/m^k$. It follows (as an exercise) that R is Noetherian if and only if R is artinian

Corollary 7.0.9: Artinian Ring Equivalent Condition I

Let R be a ring. Then R is Artinian if and only if R is Noetherian and $\dim(R) = 0$

Proof:

We have shown that R is Artinian implies R is Noetherian, and all prime ideals are maximal implies $\dim(R) = 0$. For the converse, by ??, there exists prime ideals $P_1, P_2, \dots, P_n$ such that

$$(0) = P_1 P_2 \cdots P_n$$

These prime ideals are all maximal since $\dim(R) = 0$. By the chinese remainder theorem, R is isomorphic to the direct product of a finite number of Noetherian rings of the form R/m^k

where m is a maximal ideal in R . By theorem 7.0.8, R/m^k is artinian, and so it follows that R is artinian.

Corollary 7.0.10: Artinian Ring Equivalent Condition II

Let k be a field and R a finitely generated k -algebra. Then R is artinian if and only if R is finitely generated as a k -module.

For artinian modules, we have a nice characterization if $R = k$ is a field

Lemma 7.0.11: Artin+Noetherian Module

Let M be a vector space over k . Then the following are equivalent:

1. M satisfies the A.C.C.
2. M satisfies the D.C.C.
3. M is finite-dimensional

Proof:

Naturally, (3) implies both (1) and (2). Conversely, assume $\dim_k(M) = \infty$, so pick some infinite collection of linearly independent vectors. Then we have an infinite strict chain:

$$\langle e_1 \rangle \subsetneq \langle e_1, e_2 \rangle \cdots$$

and an infinite descending chain:

$$\langle e_1, e_2, \dots \rangle \supsetneq \langle e_2, e_3, \dots \rangle \supseteq \cdots$$

completing the proof

Example 7.0.12: Artinian Rings

1. Let $n > 1$ be an integer, and let $R = \mathbb{Z}/n\mathbb{Z}$. Since R is finite, it must be Artinian. If $n = p_1^{\alpha_1} \cdots p_n^{\alpha_n}$, for some distinct primes p_n and $\alpha_i \in \mathbb{N}_{>0}$, then by the ftofgag:

$$\mathbb{Z}/n\mathbb{Z} \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_n^{\alpha_n}\mathbb{Z})$$

where each $\mathbb{Z}/p_i^{\alpha_i}\mathbb{Z}$ is a local artinian ring with maximal idea $p_i/p_i^{\alpha_i}$. The jacobson radical of this ring is $(p_1 p_2 p_n)$.

2. Let R be a finite dimensional k -algebra over some field k (it need not be commutative!). Then R is an Artinian ring, since any descending chain of R is a descending chain of k -subspces of R , which is bounded by $\dim_k(R)$. Note that if R is not commutative, R is not the finite direct product of local artinian rings, but the finite product of indecomposable artinian rings (i.e. artinian rings which cannot be split into direct sum of submodules).
3. Let $k[x]$ be the polynomial ring over a field k and $f \in k[x]$ be a nonzero polynomial. Then if $f(x) = f_1^{n_1}(x) \cdots f_k^{n_k}(x)$, by the CRT we have:

$$k[x]/(f(x)) \cong k[x]/(f_1^{n_1}(x)) \times \cdots \times k[x]/(f_k^{n_k}(x))$$

which is artinian, since $k[x]/(f(x))$ is a finite dimensional k -module (each $k[x]/(f_i^{n_i}(x))$ is a local artinian ring, with maximla ideal $(f_i(x))/(f_i^{n_i}(x))$). If $n_i = 1$ for all i , $k[x]/(f(x))$ is a product of fields, and if furthermore there was only 1 n_i (i.e. $i \in \{1\}$), then $k[x]/(f(x))$ would be a field extension

We shall give names to modules that satisfy both the Artinian and Noetherian condition:

Definition 7.0.13: Modules of Finite Length

Let M be an R -module that is Noetherian and Artinian. Then M is an R -module of *finite length*.

By proposition 3.0.6, each such module has a unique $\ell(M) \in \mathbb{N}$. It can be shown That the Jordan-Hölder Theorem applies to this class of objects. The Jordan-Hölder progrma will be dependend on the ring-action:

Example 7.0.14: Module of Finite Length

The simplest example would be when $R = k$ is a field, which gives vector-spaces. In this case, for any V

$$\ell(V) = \dim_k(V)$$

Proposition 7.0.15: Finite Length Additive Under Extensions

Let

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

Be a short exact sequence of finite length modules. Then:

$$\ell(M) = \ell(M') + \ell(M'')$$

Proof:

exercise

Exercise 7.0.1

1. Let M be a Noetherian and Artinian A -module, $\varphi : M \rightarrow M$ an A -module homomorphism. Show that there exists an $n \geq 0$ such that $M = \ker(\varphi^n) \oplus \text{im}(\varphi^n)$.
2. Let M be an R -module. Then M is of finite length iff it has a composition series.